Strategic Under-Specification: Optimal Clarification Standards for AI Assistants

Pranjal Agarwal
BITS Pilani, Hyderabad Campus

Abstract

Users of AI coding assistants decide how much of a task to specify, and the assistant decides whether to guess or to ask about what is left out. We model this as a Stackelberg game in which the assistant, whose objective places an excess cost on clarifying questions, commits to a clarification policy and users choose how much to specify. At a fixed policy, better guessing shortens prompts but never hurts users. Users who overestimate the assistant's willingness to ask under-specify, and undetected errors make this self-confirming. A uniform policy under asking friction suppresses clarification and can make better guessing lower welfare. Our main result concerns an assistant that conditions on how complete a prompt is, facing a continuum of users. Under moderate bias the optimal policy is a minimum-specification standard: ask about every open attribute once a prompt states at least $\bar{m}^*$ requirements, where $\bar{m}^*$ is the assistant's preferred prompt for the average user it pools. Diligent users are unaffected, terse users pool at the standard, and no attribute is left to a guess. We give a necessary and sufficient condition for this policy to be optimal. The welfare cost of bias is third order in the bias, against second order in a two-type model, and the mass point the standard creates in prompt lengths identifies the assistant's otherwise unobservable bias. Competition disciplines clarification only when ask rates are disclosed; without disclosure it drives clarification to zero. We derive testable predictions and an experimental design.

Keywords: Human–AI Interaction, Clarification Policies, Stackelberg Games, Contract Theory, Adverse Selection, Delegation.

Code Availability: Simulation suite and verification scripts are available at `https://github.com/pranjal-0802/strategic-underspecification`.

AI-Use Disclosure: AI assistance tools were used for simulation code, for deriving and checking some parts of the analysis. All analytical interpretations were developed, derived, and verified by the author.

1 Introduction

A user typing a coding request faces a fundamental economic trade-off: spend cognitive effort articulating edge cases, naming conventions, and constraints, or leave them implicit and trust the assistant to infer them. The assistant, in turn, must decide for each unspecified detail whether to guess or to ask. Neither choice is made in a vacuum: the user's specification effort depends on what they expect the assistant to do with silence, and the assistant's ask/guess policy is chosen (explicitly, or as an emergent property of training and product incentives) with some implicit model of how users will respond.

This paper provides a mechanism-design theory of endogenous clarification in AI assistance. Throughout, we distinguish the length of a prompt, the number m of attributes the user specifies, from the task's *under-specification*, the expected number of attributes that are neither specified by

the user nor clarified by the assistant, $(1-a)(k-m)$ when the assistant asks about each residual attribute with probability a. The two can move in opposite directions. When a biased assistant withholds clarification, sophisticated users write *longer* prompts than at the first best, yet users who do not specify everything still leave $(1-a)(k-m) > 0$ attributes to guesses, whereas at the first best ($a = 1$) none are left. *Strategic under-specification* refers to this equilibrium residual, which arises from the user's choice of m (Section 3) and from the assistant's strategic choice of a (Sections 4–7).

The paper makes four contributions.

Users. Shorter prompts are not a symptom of inefficiency in themselves: at a fixed policy, users who specify less because the assistant guesses better are better off (Proposition 1). Under-specification becomes costly when users expect more clarification than they get. Naive users then lose an amount quadratic in their belief error (Proposition 2), and when wrong guesses go undetected, learning from experience entrenches the error (Lemma 1).

A uniform policy. An assistant that commits to one ask rate for everyone asks less as its excess cost of questions rises (Corollary 2), and better guessing can then lengthen prompts and lower welfare, because the assistant withdraws clarification faster than users gain from accuracy (Proposition 5).

Screening. An assistant that conditions on prompt completeness screens users by their specification cost, without monetary transfers. With two types, the terse type receives less clarification than the assistant would otherwise give it, and the usual rent pattern reverses (Proposition 7). The main result concerns a continuum of users (Theorem 1). The leader's problem is screening with a capped in-kind transfer that the principal values, which differs from standard screening with money. Under moderate bias it is solved by a minimum-specification standard. We give a necessary and sufficient condition for its optimality, show that the welfare cost of bias is third order (Corollary 7), and show that the resulting mass point in prompt lengths identifies the bias (Corollary 8). Two features of the two-type model, effort dictation and asking distortion under moderate bias, do not survive with a continuum (Proposition 9 and Theorem 1).

Markets. Competition between assistants raises clarification to the level users would choose only if ask rates are disclosed; otherwise clarification is a shrouded attribute and competition drives it to zero (Proposition 10).

Section 2 sets up the model and the benchmark with an exogenous policy, Section 3 studies users' beliefs, and Sections 4–7 analyze pooling and two-type screening. Section 8 solves the continuum problem, Section 9 adds competition, and Sections 10–11 derive predictions and an experimental design. Section 12 examines robustness to the exact conjunctive payoff. Proofs not in the text are in the appendices.

1.1 Related literature

This work bridges contract theory and the emerging literature on formal models of human–AI interaction.

Contract theory and screening without transfers. Our screening framework builds on classical adverse-selection and mechanism design models (Mirrlees, 1971; Myerson, 1979, 1981; Baron and Myerson, 1982; Mussa and Rosen, 1978; Laffont and Tirole, 1986) that exploit the Spence–Mirrlees single-crossing property (Spence, 1973) to separate heterogeneous agents. However, unlike classical monopoly or regulatory screening where the principal uses monetary transfers to extract information rents, here the assistant screens strictly through an interactive service margin (the ask rate a offered in response to specification effort m). This connects to non-monetary screening

and costly verification (Townsend, 1979; Ben-Porath et al., 2014), where information revelation is governed by operational friction rather than cash transfers. The continuum problem of Section 8 differs from both in that the instrument is a transfer the principal is glad to make but cannot make in unlimited amounts: the cap on clarification, not rent extraction, drives the distortion, and the solution bunches users at a standard much as kinked tax schedules bunch taxpayers (Saez, 2010). The competition result applies the logic of shrouded attributes (Gabaix and Laibson, 2006) to clarification.

Delegation, authority, and non-monetary mechanism design. Our analysis relates closely to delegation and non-monetary mechanism design (Holmström, 1984; Dessein, 2002; Alonso and Matouschek, 2008; Armstrong and Vickers, 2010; Ambrus and Egorov, 2017), where a principal designs an optimal decision menu without monetary transfers, and to models of formal versus real authority (Aghion and Tirole, 1997), where an agent's upstream effort is shaped by who retains control over downstream execution. In our setting, leaving attributes unspecified implicitly delegates their resolution to the assistant. Furthermore, because the assistant's asking policy determines which state dimensions are actively resolved, our framework shares conceptual foundations with Bayesian persuasion and strategic information design (Kamenica and Gentzkow, 2011), while focusing on the effort-substitution trade-off in prompt articulation.

Costly communication and strategic disclosure. The user's specification decision formalizes costly disclosure under strategic interpretation, connecting to strategic information transmission (Crawford and Sobel, 1982), voluntary quality disclosure (Grossman, 1981), and rational inattention (Sims, 2003). In our setting, under-specification arises not from an incentive to deceive, but from the user's rational economizing on specification costs when anticipating automated assistance.

Assistance games and human–AI interaction. Our bi-level formulation models the interaction as a Stackelberg game (von Stackelberg, 1934), connecting conceptually to the literature on assistance games and cooperative inverse reinforcement learning (Hadfield-Menell et al., 2016), as well as incomplete contracting in AI alignment (Hadfield-Menell and Hadfield, 2019), where an autonomous assistant acts to maximize human utility under incomplete specification. However, while assistance games formalize the assistant's uncertainty over the human's underlying *objective or reward function*, in our setting the economic task objective is known and shared (downstream success value V and error loss L). The user privately knows the attribute values θ, but these are revealed truthfully when specified or asked about; the asymmetry that the assistant's policy must screen is the user's private *specification cost* κ. The strategic margin in our model is therefore not reward learning or preference alignment, but rather the contract-theoretic trade-off between prompt specification effort and automated clarification under costly communication. Our model provides a contract-theoretic microfoundation for this ask-versus-guess margin under asymmetric specification costs.

Clarification, ambiguity, and RLHF incentives. A parallel literature in natural language processing investigates algorithms for selective clarification under ambiguity (Rao and Daumé III, 2018; Aliannejadi et al., 2019; Zamani et al., 2020; Min et al., 2020; Kuhn et al., 2023). Recent evaluations show that state-of-the-art language models often guess prematurely under ambiguity rather than asking clarifying questions (Laban et al., 2025). Closest to our setting, Vijayvargiya et al. (2025) build an underspecified variant of SWE-Bench Verified and find that coding agents rarely detect missing information unprompted but gain substantially when they interact with the

user; Edwards and Schuster (2026) study when coding agents should ask rather than assume under uncertainty; and Yang et al. (2025) document that models infer many unstated requirements on their own but that underspecified prompts are fragile across model and prompt changes. These studies measure the ask-versus-guess margin empirically for a fixed prompt; our model treats the prompt itself as the user's equilibrium response to that margin. In conversational AI, this premature guessing is reinforced by reinforcement learning from human feedback (RLHF) and direct preference optimization (DPO), which systematically favor immediate answers and introduce conversational length and sycophancy biases (Sharma et al., 2023; Singhal et al., 2023). Existing engineering studies treat prompts as exogenous static benchmarks. Our framework endogenizes prompt specification effort as an equilibrium best response to provider clarification policies and asking frictions.

2 Environment and the Decision-Theoretic Benchmark

2.1 Setup

A task has k binary, independent, intent-relevant attributes $\theta = (\theta_1, \ldots, \theta_k) \in \{0,1\}^k$, each drawn i.i.d. uniform and privately known to the user. The user chooses a specification effort level $m \in \{0, \ldots, k\}$ and truthfully reveals m attributes.[1] Specifying m attributes costs $c(m)$, with $c(0) = 0$, $c' \geq 0$, $c'' \geq 0$.

For each of the $k-m$ unspecified attributes, the assistant either *asks* (revealing the true value, at a cost to the user) or *guesses*, succeeding independently with probability $g \in (0,1)$. An uninformed guess on a uniform binary attribute succeeds with probability $1/2$; $g > 1/2$ reflects context the assistant can exploit (repository conventions, priors over common choices), and $g < 1/2$ arises for non-binary attributes. We model the assistant's policy at this stage as a uniform ask rate $a \in [0,1]$ applied to every unspecified attribute, so each is correctly resolved with probability $q(a,g) = a + (1-a)g$. Task success requires *all* k attributes to be correctly resolved (conjunctive: one wrong parameter or convention typically breaks the deliverable). The user's payoff is

$$U(m; a, g) \;=\; V\, q(a,g)^{\,k-m} \;-\; c(m) \;-\; a\, c_Q\, (k-m),$$

with $V > 0$ the value of success and $c_Q > 0$ the user's cognitive cost of answering one clarifying question. We treat m as continuous on $[0,k]$ for tractability.

2.2 Capability improvements and effort substitution

Proposition 1. *Fix a. Let $U^*(g) = \max_{m \in [0,k]} U(m; a, g)$. Then U^* is weakly increasing in g.*

Proof. For any fixed $m \in [0,k]$, the user's objective $U(m; a, g) = Vq(a,g)^{k-m} - c(m) - ac_Q(k - m)$ is pointwise non-decreasing in g because $\partial U/\partial g = V(k-m)q(a,g)^{k-m-1}(1-a) \geq 0$. As the upper envelope of a family of weakly increasing functions, the maximized utility $U^*(g) = \max_{m \in [0,k]} U(m; a, g)$ is weakly increasing in g directly, without requiring differentiability. Under differentiability, the envelope theorem confirms that at the interior optimum $m^*(g)$,

$$\frac{dU^*}{dg} \;=\; \left.\frac{\partial U}{\partial g}\right|_{m = m^*(g)} \;=\; V(k - m^*)q^{k-m^*-1}(1-a) \;\geq\; 0.$$

[1] We assume truthful revelation conditional on the choice to specify: the strategic margin is *how much* to say, not whether to lie about what is said. A cheap-talk layer on top of this costly-specification layer is a natural extension; see Section 12.

The envelope theorem lets us ignore dm^*/dg: because m^* is chosen optimally, its first-order response to g has no first-order effect on U^*. □

Assumption 1 (Regularity)**.** *In the relevant range of m, $(k-m)\ln q(a,g) \geq -1$.*

Remark 1 (Interpreting Assumption 1 and Model Concavity)**.** *The regularity condition $(k-m)\ln q(a,g) \geq -1$ provides a sufficient condition for decreasing differences in (m,g), which guarantees monotone comparative statics via Topkis's theorem (Topkis, 1978, 1998; Milgrom and Shannon, 1994). The direction is pinned down by the cross-partial of utility: for fixed a,*

$$\frac{\partial^2 U}{\partial m\,\partial g} \;=\; -V(1-a)\,q^{\,k-m-1}\Big[(k-m)\ln q + 1\Big].$$

Thus U exhibits decreasing differences in (m,g) whenever $(k-m)\ln q + 1 \geq 0$, which is precisely Assumption 1. Because this condition is most demanding at $m=0$, decreasing differences on the entire domain $[0,k]$ (the hypothesis of Topkis's theorem) requires $k\ln q \geq -1$. When k is large or baseline accuracy is modest (e.g., $k=10, g=0.5$), this global condition is violated. A weaker local statement is available: at an interior optimum m^* satisfying the second-order condition $\partial^2 U/\partial m^2 < 0$, the implicit function theorem gives $dm^*/dg = -U_{mg}/U_{mm}$, whose sign equals that of U_{mg} evaluated at m^*. The local condition $(k-m^*)\ln q \geq -1$ therefore suffices for $dm^*/dg \leq 0$ along a smooth branch of optima. Users who articulate most attributes upstream keep $k-m^*$ small, and the local condition holds in 89.9% of our grid configurations (Section 12). Where it fails, U exhibits increasing differences at m^*, and higher guessing accuracy induces higher *specification effort: monotonicity breaks in 96.7% of the grid intervals where the local condition fails, so the condition is close to necessary as well as sufficient.*

We also note that in the exact conjunctive model, the gross benefit Vq^{k-m} is strictly convex in m (since $\partial^2[Vq^{k-m}]/\partial m^2 = Vq^{k-m}(\ln q)^2 > 0$). Under quadratic cost $c(m) = \kappa m^2/2$, U is strictly concave in m on $[0,k]$ if and only if $\kappa \geq V(\ln q)^2$ (the supremum of $Vq^{k-m}(\ln q)^2$, attained at $m=k$); otherwise the optimum may jump between local maxima, and the local argument above applies only along each smooth branch. Under the linear-risk specification of Section 2.3, by contrast, gross benefit is linear in m, guaranteeing that U is globally strictly concave in m for all $\kappa > 0$.

Corollary 1. *If Assumption 1 holds for all $m \in [0,k]$ and all g in the range considered, the set of maximizers $m^*(g)$ is weakly decreasing in g in the strong set order (Topkis's theorem). If it holds only at an interior optimum satisfying the second-order condition, $m^*(g)$ is locally weakly decreasing along that branch.*

Proof. Immediate from the cross-partial computed in Remark 1: Assumption 1 is exactly the condition $\partial^2 U/\partial m\,\partial g \leq 0$. On all of $[0,k]$ this is decreasing differences in (m,g), and Topkis's theorem applies; at a single interior optimum, the implicit function theorem gives $dm^*/dg = -U_{mg}/U_{mm} \leq 0$. □

Thus, holding the assistant's policy fixed, users specify less as the assistant becomes more adept at guessing, yet they are never worse off for doing so. Proposition 1 serves primarily as a baseline: it applies the envelope theorem to a single agent optimizing against a technology parameter that enters the payoff with a non-negative derivative, a property that holds generally in models of this class. Its purpose is not to present a surprising result, but to establish the rigorous benchmark against which claimed welfare loss from prompt shortening must be evaluated, ensuring that Proposition 2 isolates the true source of strategic inefficiency. This frames the central question of this paper: what happens once the policy a is no longer exogenous, but chosen by a strategic assistant that anticipates the user's response?

2.3 The linear-risk specification

Most of the analysis uses a linearized version of this payoff. Let specification cost be quadratic, $c(m;\kappa) = \kappa m^2/2$, where $\kappa > 0$ is the user's privately known cost type, and let L be the loss from one mis-resolved attribute. The linear-risk payoff is

$$U(m, a; \kappa) \;=\; V \;-\; \Lambda(1-a)(k-m) \;-\; \frac{\kappa}{2}m^2 \;-\; a\,c_Q\,(k-m), \qquad \Lambda \;\equiv\; L(1-g),$$

where Λ is the expected loss from leaving one attribute to a guess. We write $s \equiv \Lambda - c_Q$ for the net value of asking about one attribute rather than guessing, and maintain $s > 0$ unless stated otherwise. With $L = V$ this payoff is the first-order expansion of Vq^{k-m} around $q = 1$; it is exact in an additive-defect environment (Section 4.1). Its cross-partials are constant, $\partial^2 U/\partial m\,\partial g = -L(1-a) \le 0$ and $\partial^2 U/\partial m\,\partial a = -s < 0$. Under linear risk, U therefore has decreasing differences in (m, g) on all of $[0, k]$, and the conclusion of Corollary 1 holds globally, including when the best response is clipped at $m = k$, without Assumption 1.

3 Sources of Strategic Under-Specification

Proposition 1 assumed the assistant's ask rate is fixed and that users correctly anticipate $q(a, g)$. Both assumptions can fail.

3.1 Cost externality

If a third party (such as a code reviewer or downstream production system) bears part of the cost of an erroneous guess, the user's private defect loss L understates the social loss, causing the user to under-specify relative to the social optimum at any fixed (a, g). Under the linear-risk specification of Section 2.3 with social defect loss $L_{\text{soc}} > L$, the gap between social and private specification effort is $(L_{\text{soc}} - L)(1-g)(1-a)/\kappa$ for interior solutions. The gap contracts as guessing accuracy g improves, because autonomous errors become less frequent, and vanishes under full clarification $(a = 1)$. While this constitutes a standard Pigouvian externality, our primary focus lies in strategic and informational friction.

3.2 Miscalibrated belief updating under silent failure

If incorrect guesses are not always detected, users infer g from experience rather than observing it directly. Suppose each wrong guess is detected with probability $d(m) \in (0, 1]$, continuous and nondecreasing in m (users who specify more also test and review more closely), and that the user naively counts undetected errors as successes. The user then perceives the error rate $(1-g)\,d(m)$, hence the stake $\hat{\Lambda}(m) = \Lambda\,d(m) \le \Lambda$, and best-responds under linear risk with

$$T(m) \;\equiv\; \min\Big\{k,\; \frac{\Lambda\,d(m)(1-a) + a\,c_Q}{\kappa}\Big\}.$$

A specification level is *self-confirming* if $m = T(m)$: the belief it generates is consistent with what that user observes.

Lemma 1 (Self-confirming under-specification)**.** *Fix $a \in [0, 1)$ and let $m^*(a) = \min\{k, [\Lambda(1-a) + ac_Q]/\kappa\}$ be the best response under correct beliefs.*

(i) A self-confirming level exists, and every self-confirming level satisfies $m^{sc} \le m^*(a)$, strictly if $m^{sc} < k$ and $d(m^{sc}) < 1$.

(ii) Starting from correct beliefs, the learning path $m_{t+1} = T(m_t)$ with $m_0 = m^*(a)$ is nonincreasing and converges to the largest self-confirming level.

(iii) If $d_1 \leq d_2$ pointwise, the largest self-confirming level under d_1 is weakly below that under d_2.

Proof. T is continuous and nondecreasing and maps $[0, k]$ into itself, so $T(m) - m$ is nonnegative at 0 and nonpositive at k, and a fixed point exists. Because $d \leq 1$, $T(m) \leq m^*(a)$ for every m, so any fixed point satisfies $m^{sc} = T(m^{sc}) \leq m^*(a)$. If $m^{sc} < k$, then $m^{sc} = [\Lambda d(m^{sc})(1-a) + ac_Q]/\kappa$, which is strictly below $[\Lambda(1-a) + ac_Q]/\kappa$ when $d(m^{sc}) < 1$ and $a < 1$; with $m^{sc} < k$ this gives $m^{sc} < m^*(a)$. For (ii), $m_1 = T(m_0) \leq m_0$, and $m_t \leq m_{t-1}$ implies $m_{t+1} = T(m_t) \leq T(m_{t-1}) = m_t$. The bounded monotone sequence converges, and by continuity its limit is a fixed point. Any fixed point m' satisfies $m' \leq m_0$, and $m' \leq m_t$ implies $m' = T(m') \leq T(m_t) = m_{t+1}$, so m' lies below the limit. For (iii), both paths start at $m^*(a)$, which does not depend on d, and $m_{t+1}^{(1)} = T_1(m_t^{(1)}) \leq T_1(m_t^{(2)}) \leq T_2(m_t^{(2)}) = m_{t+1}^{(2)}$ by induction, so the limits are ordered. □

Because true utility is strictly concave in m with peak at $m^*(a)$, a self-confirming user loses at least $\frac{\kappa}{2}(m^*(a) - m^{sc})^2$. The loop amplifies the initial error: shorter prompts lower detection, which raises perceived accuracy, which shortens prompts further. Part (ii) is the declining-effort prediction P5, and part (iii) says that users who catch fewer errors settle at shorter prompts.

3.3 Belief about the assistant's policy and equilibrium under-specification

When the clarification policy a is endogenous, the user's specification decision depends on their belief regarding the assistant's asking rate. Let users be *sophisticated* (correctly inferring the assistant's true operational ask rate a_A) or *naive* (believing the assistant follows a higher clarification rate $a^\dagger > a_A$, such as the unconstrained welfare-optimal policy).

Definition 1 (Effort Substitution and Belief-Driven Under-Specification)**.** *A user interacting with an automated assistant modifies upstream specification effort m across two distinct economic margins:*

(i) ***Rational effort substitution:*** *$m^*(a) \leq m^*(0)$, where the user economizes on prompt articulation by delegating attribute resolution to automated assistance; and*

(ii) ***Belief-driven under-specification:*** *$m^*(\hat{a}) < m^*(a)$, where a user holding optimistic or miscalibrated beliefs $\hat{a} > a$ under-specifies relative to the assistant's true operational ask rate.*

In human–AI interaction, varying the clarification rate a engages two opposing economic channels:

1. *The accuracy channel:* A higher ask rate increases effective resolution accuracy $q(a, g) = a + (1-a)g$. Under decreasing differences (Assumption 1), greater accuracy reduces the marginal value of upstream prompt effort, pushing specification downward.

2. *The question-cost channel:* Each clarifying question imposes an interactive answering cost c_Q on the user. Because the total answering cost $ac_Q(k-m)$ is decreasing in specification m, a higher ask rate increases the marginal penalty of leaving attributes unspecified, pushing specification upward (pre-emption).

Under the linear-risk specification analyzed in Sections 4–6, where the net benefit of clarification $s \equiv \Lambda - c_Q > 0$ is positive, the cross-partial derivative of user utility is:

$$\frac{\partial^2 U}{\partial m \, \partial a} \;=\; -(\Lambda - c_Q) \;<\; 0.$$

The accuracy channel strictly dominates the question-cost channel, ensuring that U exhibits strictly decreasing differences in (m, a).

Proposition 2 (Prompt Under-Specification under Naive Beliefs)**.** *Under the linear-risk specification with $\Lambda > c_Q$, naive users choose weakly lower specification effort than sophisticated users:*

$$m^*_{naive} \;\leq\; m^*_{soph},$$

holding strictly ($m^*_{naive} < m^*_{soph}$) for all interior types satisfying $\kappa > \frac{\Lambda(1-a^\dagger)+a^\dagger c_Q}{k}$. Consequently, naive users achieve weakly lower true welfare, $U(m^*_{naive}; a_A) \leq U(m^*_{soph}; a_A)$, strictly so on the interior domain. Holding raw guessing accuracy g fixed, naive users under-specify relative to the assistant's true operational policy and suffer an equilibrium welfare loss.

Proof. Under linear risk on the physical domain $m \in [0, k]$, the follower's best response to an anticipated ask rate a is $m^*(\kappa; a) = \min\{k, \beta(\kappa; a)\}$, where $\beta(\kappa; a) \equiv \frac{\Lambda(1-a)+ac_Q}{\kappa}$. Since $\Lambda > c_Q$, the unconstrained response $\beta(\kappa; a)$ is strictly decreasing in a: $\partial\beta/\partial a = (c_Q - \Lambda)/\kappa < 0$. A naive user anticipating $a^\dagger > a_A$ therefore satisfies $\beta(\kappa; a^\dagger) < \beta(\kappa; a_A)$. For low-cost types with $\kappa \leq \frac{\Lambda(1-a^\dagger)+a^\dagger c_Q}{k}$, we have $\beta(\kappa; a^\dagger) \geq k$, so $\beta(\kappa; a_A) > k$ as well. Both beliefs therefore yield boundary saturation: $m^*_{\text{naive}} = m^*_{\text{soph}} = k$, and true welfare is identical ($U(m^*_{\text{naive}}; a_A) = U(m^*_{\text{soph}}; a_A)$). For all interior types satisfying $\kappa > \frac{\Lambda(1-a^\dagger)+a^\dagger c_Q}{k}$, we have $\beta(\kappa; a^\dagger) < k$, which guarantees:

$$m^*_{\text{naive}} \;=\; \beta(\kappa; a^\dagger) \;<\; \min\{k, \beta(\kappa; a_A)\} \;=\; m^*_{\text{soph}}.$$

Because true utility $U(m; a_A)$ is strictly concave in m with unique global maximizer on $[0, k]$ at m^*_{soph}, any choice $m \neq m^*_{\text{soph}}$ incurs a strict utility loss. On the fully interior domain where $m^*_{\text{soph}} = \beta(\kappa; a_A) < k$, evaluating the quadratic shortfall directly yields:

$$U(m^*_{\text{soph}}; a_A) - U(m^*_{\text{naive}}; a_A) \;=\; \frac{\kappa}{2}\big(m^*_{\text{soph}} - m^*_{\text{naive}}\big)^2 \;=\; \frac{(\Lambda - c_Q)^2}{2\kappa}\,(a^\dagger - a_A)^2 \;>\; 0,$$

confirming that interior naive users are strictly worse off.

Notice that on the interior domain, the under-specification gap is:

$$m^*_{\text{soph}} - m^*_{\text{naive}} \;=\; \frac{(\Lambda(g) - c_Q)(a^\dagger - a_A)}{\kappa} \;=\; \frac{(L(1-g) - c_Q)(a^\dagger - a_A)}{\kappa}.$$

Differentiating with respect to raw guessing accuracy g, holding the policies $(a^\dagger, a_A)$ fixed, yields:

$$\frac{\partial(m^*_{\text{soph}} - m^*_{\text{naive}})}{\partial g} \;=\; -\,\frac{L(a^\dagger - a_A)}{\kappa} \;<\; 0.$$

As the assistant's autonomous guessing capability improves, the distortion gap between naive and sophisticated users strictly contracts, because guessing errors become less frequent and the penalty for misplaced trust declines. (When a_A itself responds to g, as in Section 5, the total derivative also includes $-(s/\kappa)\, da_A/dg$, which can have either sign.) □

Remark 2 (Linear Risk Foundation and Non-Linear Cross-Partial Reversal)**.** *The monotone under-specification result of Proposition 2 relies directly on the linear-risk structure, where the cross-partial derivative is identically* $\partial^2 U/\partial m\,\partial a = -(\Lambda - c_Q) < 0$*. Under linear risk, whenever clarification is cost-effective* ($\Lambda > c_Q$)*, user utility exhibits decreasing differences globally, ensuring that anticipated asking strictly reduces prompt effort* ($m^*(\hat{a}) < m^*(a)$ *for* $\hat{a} > a$)*. In contrast, in the exact conjunctive model* $U(m,a) = Vq(a,g)^{k-m} - c(m) - ac_Q(k-m)$*, the cross-partial derivative is:*

$$\frac{\partial^2 U}{\partial m\,\partial a} \;=\; -V(1-g)\,q^{k-m-1}\big[(k-m)\ln q + 1\big] \;+\; c_Q.$$

Here, varying the ask rate activates two opposing forces: the accuracy benefit $-V(1-g)q^{k-m-1}[(k-m)\ln q + 1] \le 0$ *and the interactive question-answering friction* $+c_Q > 0$*. When question costs are large relative to task stakes or when autonomous accuracy is high, the question-cost channel dominates. For example, take* $k = 5, V = 10, \kappa = 0.5, g = 0.8, c_Q = 2$*, so that the linearized stake* $V(1-g) = 2.0 = c_Q$ *and the first-order net asking surplus is zero. Asking then has negative net value for the user (optimized utility* 4.16 *at* $a = 0$ *versus* 4.08 *at* $a = 0.1$*), and a naive user who believes* $a^\dagger = 0.6$ *while the true policy is* $a_A = 0.1$ *chooses* higher *upstream effort* ($m^*_{naive} = 3.92 > 3.15 = m^*_{soph}$) *to pre-empt conversational interruptions. Proposition 2 thus isolates the economic environment where clarification provides positive net value* ($\Lambda > c_Q$)*, which represents the canonical operating regime for helpful automated assistants.*

This identifies the dual nature of specification distortions across informational regimes:

1. *Opaque policies and strategic under-specification:* When users naively trust the assistant to provide first-best clarification ($a^\dagger = a^{FB} = 1$), they write concise prompts calibrated to full assistance ($m^*_{\text{naive}} = m^{FB} = c_Q/\kappa$). If the assistant is a biased Stackelberg leader implementing an asking-suppressed policy $a_A = a^{SE} < 1$, naive users under-specify relative to what the biased environment demands ($m^*_{\text{naive}} < m^*_{\text{soph}}$). Their welfare shortfall from misplaced trust is:

$$\Delta U_{\text{naive}} \;\equiv\; U(m^*_{\text{soph}}; a^{SE}) \;-\; U(m^*_{\text{naive}}; a^{SE}) \;=\; \frac{(\Lambda - c_Q)^2}{2\kappa}\big(1 - a^{SE}\big)^2 \;=\; \frac{s^2(1-a^{SE})^2}{2\kappa},$$

   which is strictly positive whenever the leader strategically withholds clarification ($a^{SE} < 1$).

2. *Transparent policies and defensive over-specification:* Under transparent Stackelberg commitment, sophisticated users know the leader's operational policy a^{SE}. Because the biased leader under-asks ($a^{SE} < 1$), sophisticated users cannot rely on clarification downstream. They compensate by defensively lengthening prompts beyond the first-best level ($m^*(a^{SE}) > m^{FB}$), substituting costly private effort for missing automated assistance. Prompts lengthen, yet the task stays under-specified: every type with $m^* < k$ still leaves $(1 - a^{SE})(k - m^*) > 0$ attributes to guesses, against none at the first best.

Sections 4–7 close the loop by modeling how the assistant endogenously chooses this clarification policy under pooling and screening governance mechanisms.

4 The Assistant's Policy as a Stackelberg Game

4.1 General formulation

The assistant (leader) commits ex ante to a policy π before meeting any user. A population of users (followers) has privately known specification-cost type $\kappa \sim F$ on $[\kappa_L, \kappa_H]$, with cost $c(m;\kappa) =$

$\kappa m^2/2$, and best-responds $m^*(\kappa;\pi) = \arg\max_m U(m;\pi,\kappa)$. The leader's own objective Π need not equal U: we write

$$\Pi_\kappa(m,a) \;=\; \mu_A\, U(m,a;\kappa) \;-\; (\lambda_A - c_Q)\, a\,(k-m),$$

where $\mu_A \in (0,1]$ discounts the leader's weight on true downstream user welfare, c_Q is the user's true cognitive cost of answering one clarifying question, and λ_A is the leader's own perceived or institutional cost per clarifying question. In commercial AI systems, misalignment along these margins arises from concrete engineering and economic forces:

1. *Asking friction bias* ($\lambda_A > c_Q$): Reinforcement learning from human feedback (RLHF) and preference optimization algorithms (such as DPO) commonly evaluate conversation turns in isolation. Reward models systematically penalize "evasive" clarifying questions in favor of immediate code generation because rating pools favor direct answers, reinforcing sycophancy and conversational turn penalties (Sharma et al., 2023; Singhal et al., 2023; Laban et al., 2025). Furthermore, multi-turn interactions increase token costs, context-window expansion, and inference latency, creating provider-side economic pressure to suppress clarification turns.

2. *Downstream welfare discounting* ($\mu_A < 1$): Turn-level user satisfaction signals (such as prompt acceptance or immediate positive feedback) capture immediate session fluency. In contrast, the true economic loss L from broken conventions, subtle edge-case errors, or security defects is typically discovered hours or days later in downstream execution, test suites, or code review. Consequently, training objectives tuned to immediate session retention tend to discount downstream correctness.

When asking friction is aligned ($\lambda_A = c_Q$), the leader's objective satisfies $\Pi_\kappa \equiv \mu_A U(\cdot;\kappa)$, so it has the same maximizers as true user welfare for any positive welfare weight $\mu_A > 0$. Because user utility U already incorporates private questioning cost $-ac_Q(k-m)$, expanding the leader's objective yields:

$$\Pi_\kappa(m,a) \;=\; \mu_A\left[V \;-\; \Lambda(1-a)(k-m) \;-\; \frac{\kappa}{2}m^2\right] \;-\; \Big(\lambda_A - (1-\mu_A)c_Q\Big)\, a\,(k-m).$$

The total effective weight multiplying clarification volume $-a(k-m)$ in Π_κ is therefore $\lambda_A - (1-\mu_A)c_Q$, which simplifies directly to λ_A in the benchmark case $\mu_A = 1$. Downstream welfare discounting $\mu_A < 1$ creates policy distortion through interaction with excess asking friction ($\lambda_A \neq c_Q$), inflating the effective friction per unit of altruism $(\lambda_A - c_Q)/\mu_A$. The leader solves $\pi^* = \arg\max_\pi \mathbb{E}_{\kappa\sim F}[\Pi_\kappa(m^*(\kappa;\pi),\pi)]$, a bilevel program.

Remark 3 (User welfare, social welfare, and the source of the wedge)**.** *The sources of $\lambda_A > c_Q$ listed above are of two kinds. Reward-model penalties on clarifying turns are a bias of the training signal; token costs, context growth, and latency are real resource costs borne by the provider. Write $\lambda_A - c_Q = c_A + \beta_A$, with $c_A \geq 0$ the provider's real cost per question and $\beta_A \geq 0$ pure bias. Social welfare is then $W_{soc} = \mathbb{E}[U - c_A\, a(k-m)]$, and under linear risk the socially efficient policy asks only if $\Lambda > c_Q + c_A$. Throughout, welfare $W \equiv \mathbb{E}[U]$ means* user *welfare. The statements that $a^{FB} = 1$, that welfare falls in Proposition 5, and the welfare ordering of Section 7 concern users, and they coincide with statements about social welfare only when $c_A = 0$. At the other extreme ($\mu_A = 1$, $\beta_A = 0$) the leader maximizes W_{soc} exactly. The Model I rate is then the socially optimal pooling rate given users' best responses, and the distortion of Proposition 7 is the constrained-efficient response of a planner who cannot observe κ and cannot use transfers. Distinguishing c_A from β_A empirically requires measuring provider costs directly.*

Remark 4 (Existence)**.** *For Model I below ($\pi = a \in [0,1]$, a compact scalar policy), $m^*(\kappa; a)$ is continuous in a by Berge's Maximum Theorem (since U is strictly concave in m on the compact domain $[0,k]$), so the expected leader objective $\mathbb{E}_\kappa[\Pi_\kappa(m^*(\kappa;a),a)]$ is continuous on $[0,1]$ and a maximizer exists by Weierstrass's Extreme Value Theorem. For Model II below (a finite menu over two types), the feasible allocation space is a non-empty, compact subset of $[0,k]^2 \times [0,1]^2$ defined by continuous (bilinear in (m,a) and quadratic in m) IC and IR constraints. The constraint set is non-empty because the zero-asking menu $\{(m_L^0, 0), (m_H^0, 0)\}$ with $m_\kappa^0 = \min\{k, \Lambda/\kappa\}$ is feasible: each type receives its own unassisted optimum, so both IC constraints hold, and IR holds for any reservation utility not exceeding the unassisted payoff. By Weierstrass's theorem, the continuous leader objective attains a global maximum on this compact set.*

For tractability, Sections 4–7 evaluate the linear-risk objective of Section 2.3:

$$U(m, a; \kappa) \;=\; V \;-\; \Lambda(1-a)(k-m) \;-\; \frac{\kappa}{2}m^2 \;-\; a\,c_Q\,(k-m),$$

where $\Lambda \equiv L(1-g)$. This specification admits two complementary economic microfoundations:

1. *Additive independent defects ($kL \le V$):* Each mis-resolved attribute independently produces a localized defect cost $L \le V/k$, so losses add up across attributes and exhaust the task value only if every attribute is defective. Our benchmark calibration sets $k = 10, L = 10, V = 100$, where full failure ($kL = 100$) exactly exhausts task value V.

2. *Systemic bottleneck conjunctive model ($L = V$):* The project succeeds only if all attributes are defect-free: $Vq(a,g)^{k-m}$. A first-order Taylor expansion around $q \to 1$ yields $Vq^{k-m} \approx V - V(1-q)(k-m)$, which corresponds to setting per-attribute stake $L = V = 100$.

Throughout Sections 4.2–6, we maintain the physical feasibility constraint $m \in [0,k]$, so that followers cannot specify more attributes than exist in the task. When unconstrained first-order conditions yield $m^*(\kappa; a) = \frac{\Lambda(1-a)+ac_Q}{\kappa} > k$ (which occurs for low-cost types with $\kappa < [\Lambda(1-a) + ac_Q]/k$, a threshold that falls from Λ/k at $a = 0$ to c_Q/k at $a = 1$), best-response effort saturates at the physical boundary $m^* = k$.

4.2 The first-best benchmark and physical feasibility

Before analyzing either strategic regime, we characterize the fully-informed, fully-aligned social optimum:

$$(m^{FB}(\kappa), a^{FB}(\kappa)) \;=\; \arg\max_{m\in[0,k],\, a\in[0,1]} U(m, a; \kappa).$$

Proposition 3 (First-Best Policy on the Feasible Domain)**.** *On the physically feasible domain $m \in [0,k]$, under the linear-risk specification with $\Lambda > c_Q$:*

(i) Asking weakly dominates guessing for every cost type $\kappa > 0$: $a^{FB}(\kappa) = 1$, with optimal specification $m^{FB}(\kappa) = \min\{k, c_Q/\kappa\}$.

(ii) For any type with $\kappa > c_Q/k$, asking strictly dominates guessing: $U(m^*(1), 1) > U(m^*(0), 0)$. For boundary types with $\kappa \le c_Q/k$, specification effort saturates at $m^{FB} = k$, leaving zero residual attributes ($k - m = 0$) and rendering the asking policy payoff-irrelevant.

Proof. For any fixed specification effort $m \in [0,k]$, the partial derivative of utility with respect to a is:

$$\frac{\partial U}{\partial a} \;=\; (\Lambda - c_Q)(k-m).$$

Because $\Lambda > c_Q$, this derivative is strictly positive for all $m < k$ and zero at $m = k$. Hence, for every $m \in [0, k]$, $a = 1$ maximizes $U(m, \cdot\,; \kappa)$ over $[0, 1]$, so $a^{FB} = 1$ and

$$m^{FB} \;=\; \arg\max_{m \in [0,k]} U(m, 1; \kappa) \;=\; \min\{k, c_Q/\kappa\},$$

proving part (i). To quantify the gain over pure guessing (part (ii)), compare the optimized payoffs under $a \in \{0, 1\}$. Under $a = 1$, the user chooses $m^*(1) = \min\{k, c_Q/\kappa\}$. Under $a = 0$, the user chooses $m^*(0) = \min\{k, \Lambda/\kappa\}$. We evaluate the three type intervals:

- If $\kappa \geq \Lambda/k$, both best responses lie in $[0, k]$: $m^*(1) = c_Q/\kappa < m^*(0) = \Lambda/\kappa \leq k$. Evaluating the welfare difference:

$$U(m^*(1), 1; \kappa) - U(m^*(0), 0; \kappa) \;=\; (\Lambda - c_Q)\left(k - \frac{\Lambda + c_Q}{2\kappa}\right).$$

  Since $\kappa \geq \Lambda/k > \frac{\Lambda + c_Q}{2k}$, the bracketed term satisfies $k - \frac{\Lambda + c_Q}{2\kappa} > k - k = 0$. Hence $U(m^*(1), 1; \kappa) > U(m^*(0), 0; \kappa)$ strictly.

- If $c_Q/k < \kappa < \Lambda/k$, unconstrained effort under $a = 0$ exceeds k, so specification under guessing saturates at the boundary $m^*(0) = k$, yielding $U(k, 0; \kappa) = V - \frac{\kappa}{2}k^2$. Under asking, effort remains interior at $m^*(1) = c_Q/\kappa < k$, yielding $U(m^*(1), 1; \kappa) = V - c_Q k + \frac{c_Q^2}{2\kappa}$. Subtracting:

$$U(m^*(1), 1; \kappa) - U(k, 0; \kappa) \;=\; \frac{\kappa k^2}{2} - c_Q k + \frac{c_Q^2}{2\kappa} \;=\; \frac{(\kappa k - c_Q)^2}{2\kappa} \;>\; 0.$$

- If $\kappa \leq c_Q/k$, specification effort saturates at $m^* = k$ under both policies, so $U(k, 1; \kappa) = U(k, 0; \kappa) = V - \frac{\kappa}{2}k^2$.

Therefore, $a^{FB}(\kappa) = 1$ is globally optimal across all types $\kappa > 0$, with indifference only on the boundary set $\kappa \leq c_Q/k$ where $m^{FB} = k$. □

Remark 5 (The Unclipped Threshold κ^* and Genuine Asking–Guessing Trade-offs)**.** *In unconstrained algebra without the feasibility constraint $m \leq k$, setting the difference $(\Lambda - c_Q)(k - \frac{\Lambda + c_Q}{2\kappa})$ to zero yields the formal threshold $\kappa^* \equiv \frac{\Lambda + c_Q}{2k}$. However, since $\Lambda > c_Q$, we have $\kappa^* < \Lambda/k$ strictly. The region $\kappa < \kappa^*$ where guessing formally appeared to beat asking in unclipped algebra lies entirely in the unphysical domain where unconstrained effort Λ/κ exceeds k (the user would specify more attributes than exist in the task). Once $m \leq k$ is enforced, asking weakly dominates guessing everywhere under linear risk because asking resolves uncertainty at lower unit cost ($c_Q < \Lambda$) while allowing the user to economize on specification effort.*

Thresholds at which a^{FB} switches between 0 *and* 1 *across the type space can arise in two natural richer specifications:*

1. Exact conjunctive success: *Under $Vq(a, g)^{k-m} - c(m) - ac_Q(k - m)$, the marginal gain from raising a at fixed m is $(k - m)[V(1 - g)q^{k-m-1} - c_Q]$. Because q^{k-m-1} rises as $k - m$ falls, the per-question value of asking is* highest *for users who leave few attributes open. Guessing can therefore be first best when the per-question stake $V(1 - g)$ is small relative to c_Q, or for terse users on long tasks, whose total question bill $c_Q(k - m)$ is comparable to V. Section 12 quantifies when this type-dependence arises.*

2. Heterogeneous attribute stakes: *If attributes vary in importance with stakes Λ_i, efficiency prescribes asking only on high-stakes attributes ($\Lambda_i > c_Q$) and guessing on low-stakes attributes ($\Lambda_i < c_Q$). If the assistant must use one ask rate for every open attribute, the first best switches from guessing to asking at a threshold in κ (Proposition 8).*

5 Model I: A Pooling (Scalar) Policy

The leader is restricted to a single population-wide ask rate $a \in [0,1]$ and cannot condition on user-specific attributes or observed effort. This provides a natural benchmark for policies embedded uniformly at training time or applied without user-level profiling.

5.1 User best response

The FOC $\partial U/\partial m = \Lambda(1-a) - \kappa m + a c_Q = 0$ gives, with $\beta(\kappa) = \Lambda/\kappa$ and $\gamma(\kappa) = (\Lambda - c_Q)/\kappa$,

$$m^*(\kappa; a) \;=\; \beta(\kappa) - a\,\gamma(\kappa), \qquad \frac{\partial m^*}{\partial a} \;=\; -\gamma(\kappa) \;=\; \frac{c_Q - \Lambda}{\kappa}.$$

(Here $\beta(\kappa)$ and $\gamma(\kappa)$ denote the intercept and slope of the best response; the two-argument $\beta(\kappa; a)$ used in the proof of Proposition 2 equals $\beta(\kappa) - a\gamma(\kappa)$.) The closed forms below assume interior specification levels $m^* \in (0, k)$ for every type and every $a \in [0,1]$, which holds when the support satisfies $\kappa_L \geq \Lambda/k$; for lower-cost types the best response saturates at $m^* = k$. The population moments $\bar{R}_0$ and $\bar{\gamma}$ are then exact. When clipping binds they are not, and we re-solve the leader's problem numerically on the clipped best responses wherever a result is stated for such a calibration (Remark 8, Appendix C). The corner property of the unbiased case survives clipping exactly (Corollary 4).

Remark 6 (Sign of the ask-rate effect on specification)**.** *Unlike the effect of guessing accuracy in Corollary 1, the sign of $\partial m^*/\partial a$ here does not depend on type or on m: it is positive for* every *type iff $c_Q > \Lambda$, i.e., iff answering a question costs users more than the expected loss it avoids. When that condition holds, a higher population ask rate induces users to specify* more*, not less; they pre-empt costly clarifying questions by articulating the answer directly. This introduces a new economic force relative to Section 2, where a operated solely through effective accuracy, and generates a distinct, directly testable prediction (prediction P1′ below).*

5.2 The leader's problem

Let $R(a, \kappa) = k - m^*(\kappa; a)$ denote the residual unspecified attributes. In the pooling regime, the assistant commits ex ante to an operational ask rate $a \in [0,1]$ applied population-wide, without observing or directly contracting on individual specification effort.

Evaluating the general leader objective from Section 4.1 at the follower's optimal best response $m^*(\kappa; a) = \beta(\kappa) - a\,\gamma(\kappa)$:

$$\Pi_\kappa(m^*(\kappa; a), a) \;=\; \mu_A\, U(m^*(\kappa; a), a; \kappa) \;-\; (\lambda_A - c_Q)\, a\, \big(k - m^*(\kappa; a)\big).$$

Substituting $U(m^*(\kappa; a), a; \kappa) = V - \Lambda(1-a)(k-m^*) - \frac{\kappa}{2}(m^*)^2 - a\, c_Q\,(k - m^*)$, using $k - m^*(\kappa; a) = R_0(\kappa) + a\,\gamma(\kappa)$, and defining the effective resolution stake $s \equiv \Lambda - c_Q$ and the assistant's excess asking friction $b \equiv \lambda_A - c_Q$, expanding in powers of a yields:

$$\Pi_\kappa(m^*(\kappa; a), a) \;=\; \mu_A \left(V - \Lambda k + \frac{\Lambda^2}{2\kappa}\right) \;+\; (\mu_A s - b)\, R_0(\kappa)\, a \;+\; \frac{(\mu_A s - 2b)\,\gamma(\kappa)}{2}\, a^2.$$

Averaging over $\kappa \sim F$ requires no moments beyond the population sufficient statistics already defined, $\bar{R}_0 = k - \mathbb{E}[\beta]$ and $\bar{\gamma} = \mathbb{E}[\gamma] = s\,\mathbb{E}[1/\kappa]$:

$$\Pi(a) \;=\; \text{const} \;+\; \alpha_1\, a \;+\; \alpha_2\, a^2,$$

where const $= \mu_A(V - \Lambda k) + \frac{\mu_A \Lambda^2}{2}\mathbb{E}[1/\kappa]$, and

$$\alpha_1 \;\equiv\; (\mu_A s - b)\,\bar{R}_0, \qquad \alpha_2 \;\equiv\; \frac{(\mu_A s - 2b)\,\bar{\gamma}}{2}.$$

Lemma 2 (Leader Objective as User Welfare with an Asking Friction Wedge)**.** *For any allocation (m, a), the leader's objective satisfies:*

$$\Pi(m, a) \;=\; \mu_A\Big[U(m, a) \;-\; \tilde{\rho}\, a\,(k - m)\Big],$$

where $\tilde{\rho} \equiv \frac{\lambda_A - c_Q}{\mu_A}$ is the normalized excess asking friction. The leader's objective incorporates true user welfare but introduces a per-question friction wedge $\tilde{\rho}$. In the leader's evaluation, clarifying questions are priced at an effective cost $c_Q^{eff} \equiv c_Q + \tilde{\rho}$. However, this is not an equivalence to a benevolent social planner, because followers continue to best-respond against their true private question cost c_Q, not c_Q^{eff}. Because $\mu_A > 0$ enters strictly as an overall positive scalar multiplier on the bracketed term, optimal leader policies depend on (μ_A, λ_A) solely through the excess friction ratio $\tilde{\rho}$.

Remark 7 (Scope of the non-identification result)**.** *Lemma 2 relies on μ_A scaling all of U, including the user's specification and question costs. The motivating story for $\mu_A < 1$, training signals that miss defects discovered downstream, suggests discounting only the defect loss:*

$$\Pi_\kappa^{alt} \;=\; V - \mu_A\Lambda(1-a)(k-m) - \frac{\kappa}{2}m^2 - \lambda_A\, a(k-m) \;=\; U + (1-\mu_A)\Lambda(1-a)(k-m) - (\lambda_A - c_Q)\, a(k-m).$$

The leader's marginal value of asking at fixed m is then $(\mu_A\Lambda - \lambda_A)(k - m)$, so discounting and friction still both push toward guessing, but they enter as separate wedges. The first scales with $\Lambda = L(1 - g)$ and the second does not, so exogenous variation in g or L separates μ_A from λ_A in principle. Non-identification is therefore a property of the multiplicative specification, which we adopt for tractability; under Π^{alt} the closed forms of Sections 5–6 would have to be re-derived.

Proposition 4 (Stackelberg-optimal pooling rate)**.** *Provided $\alpha_2 < 0$ (the second-order condition for strict concavity, equivalent to $2(\lambda_A - c_Q) > \mu_A(\Lambda - c_Q)$), the leader's unconstrained stationary point is*

$$a^{SE} \;=\; -\frac{\alpha_1}{2\alpha_2} \;=\; -\frac{(\mu_A s - b)\,\bar{R}_0}{(\mu_A s - 2b)\,\bar{\gamma}}.$$

When this stationary point lies in $[0, 1]$, it is the unique global maximum of $\Pi(a)$ on $[0, 1]$.

Figure 1 compares this closed form with a numerical grid search over a.

Corollary 2 (Friction bias suppresses the pooling ask rate on the interior branch)**.** *On the interior branch $\alpha_2 < 0$, differentiating the closed form in Proposition 4 directly with respect to λ_A (holding primitives $\mu_A, \Lambda, c_Q, \bar{R}_0, \bar{\gamma}$ fixed) gives:*

$$\frac{\partial a^{SE}}{\partial \lambda_A} \;=\; -\frac{\bar{R}_0}{\bar{\gamma}}\,\frac{\mu_A s}{(\mu_A s - 2b)^2} \;=\; -\frac{\bar{R}_0}{\bar{\gamma}}\,\frac{\mu_A(\Lambda - c_Q)}{\big[\mu_A(\Lambda - c_Q) - 2(\lambda_A - c_Q)\big]^2}.$$

For populations supported on the interior specification domain ($\kappa \geq \Lambda/k$ for all types in the support), follower effort satisfies $m^*(\kappa; 0) = \Lambda/\kappa \leq k$, ensuring that expected residual unspecified attributes

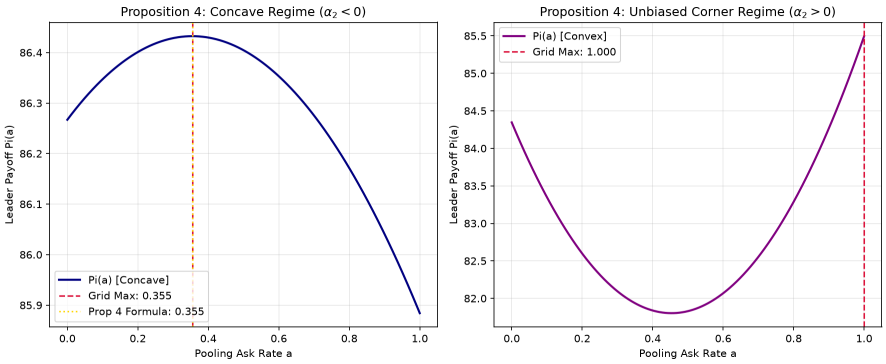


Figure 1: Leader pooling payoff $\Pi(a)$ and equilibrium ask rate a^{SE} under Model I, comparing the unclipped closed form against numerical grid optimization.

under silence satisfy $\bar{R}_0 = k - \mathbb{E}[\beta] \geq 0$*, strictly unless every type sits exactly at* $\kappa = \Lambda/k$*. Because* $\bar{\gamma} > 0$*,* $\mu_A > 0$*, and* $s > 0$*, it follows that whenever* $\bar{R}_0 > 0$*:*

$$\frac{\partial a^{SE}}{\partial \lambda_A} \; < \; 0.$$

On this interior domain, higher perceived clarification friction λ_A *strictly suppresses the assistant's equilibrium asking rate. A clipping-robust weak version follows from a single-crossing argument:* $\partial \Pi_\kappa(m^*(\kappa; a), a)/\partial \lambda_A = -a(k - m^*(\kappa; a))$*, and* $a(k - m^*(\kappa; a))$ *is non-decreasing in* a *(because* m^* *is non-increasing in* a *when* $s > 0$*, with or without clipping). Hence* $\Pi(a)$ *has decreasing differences in* (a, λ_A)*, and by Topkis's theorem the set of optimal pooling rates is weakly decreasing in* λ_A *for any type distribution, including populations with clipped low-cost types. Figure 2 plots* a^{SE} *against* λ_A *for several values of* μ_A*.*

Corollary 3 (Continuous transition of the pooling ask rate across friction regimes)**.** *Under asking friction bias (*$\lambda_A > c_Q$*), the second-order coefficient satisfies* $\alpha_2 < 0$ *if and only if* $\tilde{\rho} > s/2$*. For a continuum population supported on the interior specification domain (*$\kappa \geq \Lambda/k$*, so that* $\bar{R}_0 > 0$*), the endpoint crossover threshold satisfies* $\tilde{\rho}^* \equiv \frac{s(\bar{R}_0 + \bar{\gamma}/2)}{\bar{R}_0 + \bar{\gamma}} > s/2$ *strictly. Consequently, on the entire convex branch (*$\tilde{\rho} \leq s/2 < \tilde{\rho}^*$*), the leader strictly prefers universal clarification (*$\Pi(1) > \Pi(0)$*), selecting* $a^{SE} = 1$ *unconditionally. On the strictly concave branch (*$\tilde{\rho} > s/2$*), the optimal pooling policy transitions continuously on* $[0, 1]$ *across three distinct friction regimes:*

1. Mild bias ($\tilde{\rho} \leq \tilde{\rho}_1$): *For* $\tilde{\rho} \leq \tilde{\rho}_1 \equiv \frac{s(\bar{R}_0 + \bar{\gamma})}{\bar{R}_0 + 2\bar{\gamma}}$*, the unconstrained stationary point satisfies* $-\alpha_1/(2\alpha_2) \geq 1$*, so the constrained maximum on* $[0, 1]$ *is universal clarification:* $a^{SE} = 1$*.*

2. Moderate bias ($\tilde{\rho}_1 < \tilde{\rho} < s$): *The unconstrained stationary point is strictly interior, so the optimal ask rate is uniquely:*

$$a^{SE} \; = \; -\frac{\alpha_1}{2\alpha_2} \; = \; \frac{(s - \tilde{\rho})\bar{R}_0}{(2\tilde{\rho} - s)\bar{\gamma}} \; \in \; (0, 1),$$

which is strictly decreasing in friction ($\partial a^{SE}/\partial \tilde{\rho} < 0$*). At the threshold value* $\tilde{\rho} = \tilde{\rho}^* \in (\tilde{\rho}_1, s)$*, endpoint payoffs satisfy* $\Pi(1) = \Pi(0)$*, which corresponds geometrically to the quadratic vertex lying exactly at* $a^{SE} = 1/2$*. The interior optimum transitions smoothly through this point without any regime discontinuity.*

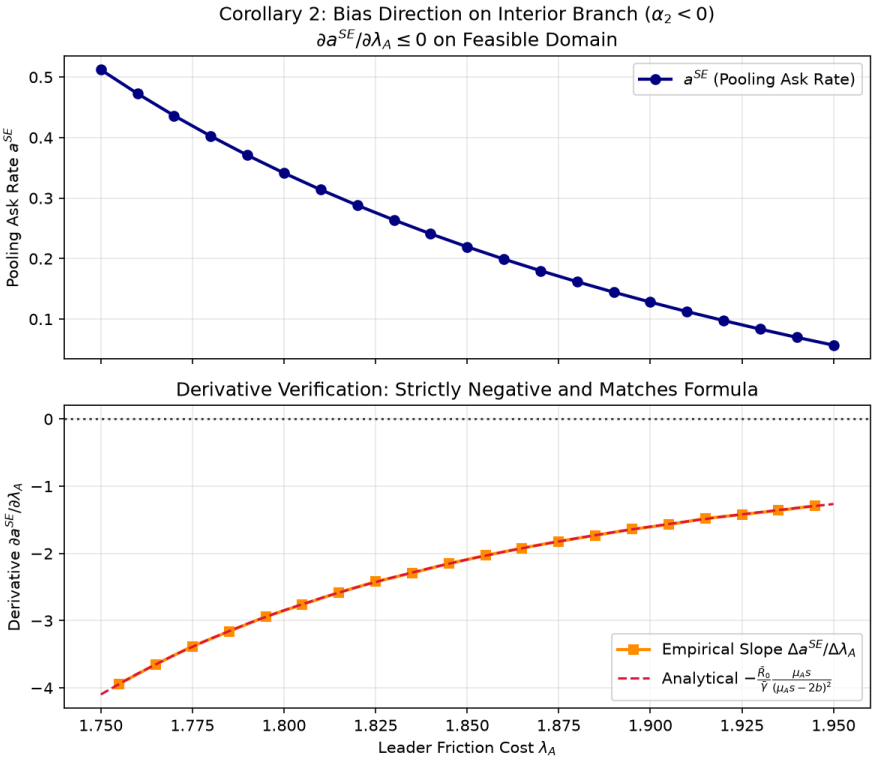


Figure 2: Perceived asking friction λ_A suppresses the Stackelberg pooling ask rate ($\partial a^{SE}/\partial \lambda_A \leq 0$) across varying altruism levels μ_A.

3. Prohibitive friction ($\tilde{\rho} \geq s$, equivalently $c_Q^{eff} \geq \Lambda$): *The leader unconstrainedly prefers zero clarification ($-\alpha_1/(2\alpha_2) \leq 0$), shutting down asking completely: $a^{SE} = 0$.*

This continuous transition characterizes the single unclipped quadratic representation with continuous type support; in discrete type distributions (such as two-type populations $\kappa \in \{\kappa_L, \kappa_H\}$) or under boundary clipping, piecewise best-response regimes can induce discrete policy shifts.

Corollary 4 (Pooling is a corner solution in the unbiased case)**.** *In the unbiased case ($\mu_A = 1, \lambda_A = c_Q \implies b = 0$), the quadratic coefficient is:*

$$\alpha_2 \;=\; \frac{s\,\bar{\gamma}}{2} \;=\; \frac{s^2\,\mathbb{E}[1/\kappa]}{2} \;=\; \frac{(\Lambda - c_Q)^2\,\mathbb{E}[1/\kappa]}{2} \;\geq\; 0 \quad \textit{always.}$$

The second-order condition for an interior maximum therefore fails: $\Pi(a)$ is weakly convex on $[0,1]$, its stationary point is a local minimum, and the true optimum is always a corner, $a^{SE} \in \{0,1\}$, not an interior compromise. Comparing the two corners directly in unclipped algebra:

$$\Pi(1) - \Pi(0) \;=\; \alpha_1 + \alpha_2 \;=\; s\left(\bar{R}_0 + \frac{\bar{\gamma}}{2}\right) \;=\; (\Lambda - c_Q)\left[k - \frac{\Lambda + c_Q}{2}\,\mathbb{E}\left[\frac{1}{\kappa}\right]\right],$$

which yields the formal unclipped threshold $k^* \equiv \frac{\Lambda + c_Q}{2}\mathbb{E}[1/\kappa]$. When follower best responses are physically constrained to the feasible domain $m \leq k$, Proposition 3 implies that asking ($a = 1$) weakly dominates guessing for every individual type in the support, so unbiased pooling under physical feasibility selects $a^{SE} = 1$ unconditionally. Under the exact conjunctive payoff, first-best asking can differ across types, but only on long tasks, with the terse type guessing (Section 12). Figure 3 contrasts the convex unbiased payoff with the concave biased one.

The corner property itself does not rely on the unclipped algebra. For each m, $U(m, a; \kappa)$ is affine in a, so the clipped value $U^*(a; \kappa) = \max_{m \in [0,k]} U(m, a; \kappa)$ is a pointwise maximum of affine functions and hence convex in a. With $\Pi = \mathbb{E}_\kappa[U^*(a; \kappa)]$ in the unbiased case, Π is convex on $[0,1]$ for any *type distribution, with or without boundary clipping, and its maximum is attained at a corner.*

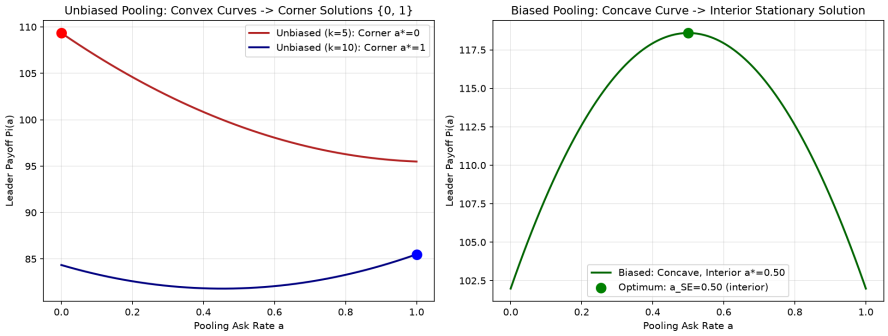


Figure 3: Payoff curvature across friction regimes: weakly convex corner solution in the unbiased regime ($\mu_A = 1, \lambda_A = c_Q$) versus strictly concave interior optimum under friction bias ($\lambda_A > c_Q$).

Proposition 5 (Equilibrium Reversal of Guessing Capability: The Guessing Trap)**.** *Under asking friction bias ($\lambda_A > c_Q$), on the concave interior branch ($\alpha_2 < 0$, $a^{SE} \in (0,1)$) with every type's best*

response interior ($m^*(\kappa) \in (0,k)$), an increase in assistant guessing capability g affects the pooling clarification rate according to:

$$\frac{da^{SE}}{dg} \;<\; 0 \iff \mu_A \bar{R}_0(g) \;>\; (\mu_A s(g) - b)\,\mathbb{E}\left[\frac{1}{\kappa}\right]\left(1 - 2a^{SE}\right).$$

In particular:

(i) On the interior branch with $\bar{R}_0 > 0$, if $a^{SE} \geq 1/2$ the condition holds automatically, so $\frac{da^{SE}}{dg} < 0$. If $a^{SE} < 1/2$, the derivative can reverse when $\bar{R}_0$ is small.

(ii) When $\frac{da^{SE}}{dg} < 0$, prompt lengthening and welfare decline are mathematically equivalent and hold simultaneously if and only if

$$-\frac{da^{SE}}{dg} \;>\; \frac{L(1 - a^{SE})}{\Lambda(g) - c_Q},$$

reversing the comparative statics of Proposition 1 in Stackelberg equilibrium:

- ***Equilibrium prompt lengthening:*** $\frac{d\mathbb{E}[m^*]}{dg} > 0$.
- ***Equilibrium welfare decline:*** $\frac{dW}{dg} < 0$.

Proof. On the interior branch, the stationary point is $a^{SE}(g) = -\frac{\alpha_1(g)}{2\alpha_2(g)} = \frac{(\mu_A s(g) - b)\bar{R}_0(g)}{(2b - \mu_A s(g))\bar{\gamma}(g)}$, where $\Lambda(g) = L(1-g)$, $s(g) = L(1-g) - c_Q$, $\bar{R}_0(g) = k - \Lambda(g)\mathbb{E}[1/\kappa]$, and $\bar{\gamma}(g) = s(g)\mathbb{E}[1/\kappa]$. Differentiating with respect to g and collecting terms in $\mu_A, b, s(g), \bar{R}_0(g)$, and $\mathbb{E}[1/\kappa]$ yields:

$$\frac{da^{SE}}{dg} \;=\; -\frac{L}{\bar{\gamma}(g)(2b - \mu_A s(g))}\left[\mu_A \bar{R}_0(g) \;-\; (\mu_A s(g) - b)\mathbb{E}\left[\frac{1}{\kappa}\right]\left(1 - 2a^{SE}\right)\right].$$

Because the prefactor $-\frac{L}{\bar{\gamma}(g)(2b - \mu_A s(g))}$ is strictly negative on the concave interior branch ($2b > \mu_A s$), it follows that $da^{SE}/dg < 0$ if and only if $\mu_A \bar{R}_0(g) > (\mu_A s(g) - b)\mathbb{E}[1/\kappa](1 - 2a^{SE})$. When $a^{SE} \geq 1/2$, $(1 - 2a^{SE}) \leq 0$. Interiority with $\bar{R}_0 > 0$ and $2b > \mu_A s$ requires $\mu_A s > b$ (otherwise $a^{SE} \leq 0$), so the right-hand side is non-positive while $\mu_A \bar{R}_0 > 0$, and the condition holds.

By the chain rule, total equilibrium follower effort satisfies:

$$\frac{dm^*}{dg} \;=\; \frac{\partial m^*}{\partial g} \;+\; \frac{\partial m^*}{\partial a}\frac{da^{SE}}{dg} \;=\; \frac{1}{\kappa}\left[-L(1 - a^{SE}) \;-\; s(g)\frac{da^{SE}}{dg}\right].$$

Similarly, by the envelope theorem on follower choice of m^*, total equilibrium user welfare $W(g) \equiv \mathbb{E}[U(m^*(g), a^{SE}(g); g)]$ satisfies:

$$\frac{dW}{dg} \;=\; \mathbb{E}\left[\left.\frac{\partial U}{\partial g}\right|_{m^*}\right] \;+\; \mathbb{E}\left[\left.\frac{\partial U}{\partial a}\right|_{m^*}\right]\frac{da^{SE}}{dg} \;=\; \mathbb{E}[(k - m^*)]\left[L(1 - a^{SE}) \;+\; s(g)\frac{da^{SE}}{dg}\right].$$

Because $\mathbb{E}[k - m^*] > 0$ and $\mathbb{E}[1/\kappa] > 0$, prompt lengthening ($d\mathbb{E}[m^*]/dg > 0$) and welfare decline ($dW/dg < 0$) are mathematically equivalent and hold simultaneously if and only if $-da^{SE}/dg > \frac{L(1-a^{SE})}{s(g)}$. □

Two forces act on a^{SE} when g rises with the wedge held fixed. Better guessing shrinks the net gain from a question, $s(g) = L(1-g) - c_Q$, while the provider's excess friction per question $\tilde{\rho}$ (latency, tokens, turn penalties) does not shrink with the model's accuracy. The friction ratio $r \equiv \tilde{\rho}/s(g)$ therefore rises, at rate $L\tilde{\rho}/s(g)^2$, toward the shutdown threshold $r = 1$ of Corollary 3, and the first factor of $a^{SE} = \frac{1-r}{2r-1} \cdot \frac{\bar{R}_0}{\bar{\gamma}}$ falls. Against this, users leave more attributes open when guessing improves, so $\bar{R}_0$ rises and $\bar{R}_0/\bar{\gamma}$ rises with it. The trap requires the first force to win by the margin in part (ii); part (i) and Remark 8 show that it need not.

Remark 8 (Analytical Conditions versus Clipped Numerical Calibration)**.** *Proposition 5 characterizes the unclipped interior branch where best responses remain interior ($m^*(\kappa) < k$). When $a^{SE} < 1/2$, the sign condition can fail, so it is not vacuous. For example, with many-valued attributes where guessing accuracy satisfies $g \in [0.05, 0.10]$ (with $k = 12, L = 17, c_Q = 4.5, \lambda_A = 11.5$, $\mu_A = 1$, and $\kappa \sim$ Uniform$[1.4, 1.5]$), all types remain unclipped and a^{SE} rises from 0.211 to 0.230 as g increases.*

In the benchmark calibration ($k = 10, L = 10, c_Q = 2, \mu_A = 1, b = 1.2, \kappa \sim$ Uniform$[0.2, 0.6]$), unclipped moments at $g = 0.5$ give $\bar{R}_0 = -3.73$ and $\alpha_2 = +2.47$, so the closed form of Proposition 4 does not apply. On the binary-attribute range $g \in [0.50, 0.65]$, low-cost followers are subject to boundary clipping $m^*(\kappa) = \min\{k, \dots\}$, and we solve the leader's problem directly on the clipped best responses. Evaluating the exact clipped model via numerical quadrature: as guessing capability improves from $g = 0.50$ to $g = 0.60$, the equilibrium ask rate collapses from $a^{SE} = 0.661$ to $a^{SE} = 0.424$. In response to this withdrawal of clarification, average prompt specification expands from $\mathbb{E}[m] = 7.728$ to 7.953, and expected user welfare declines from $W = 82.134$ to 81.843. To gauge how common the trap is, we draw 2,000 random parameter vectors ($k \in \{6, \dots, 15\}$, $g \in [0.5, 0.8]$, $\tilde{\rho} \in [0, 1.5\,s]$, uniform type distributions) and re-solve the clipped pooling equilibrium at g and $g + 0.02$. Simultaneous prompt lengthening and welfare reduction occur in 12.2% of all draws and in 51.6% of the 318 draws with an interior a^{SE}. These frequencies depend on the sampling distribution and should be read as evidence that the trap is not a knife-edge, not as population estimates.

The magnitude is modest: in the benchmark, welfare falls by 0.29, or 0.35% of W. The comparative static also holds the wedge $\tilde{\rho}$ fixed as g changes. The trap is an instance of the general second-best principle that improving one margin can lower welfare when another margin is distorted. What is specific to this setting is the channel: the assistant withdraws clarification as it guesses better, and users offset part of the withdrawal with longer prompts. When part of the wedge is a real provider cost, the user-welfare decline is partly offset by the provider's savings (Remark 3).

This structure matters for the comparison in Section 7: under linear risk, Model I falls short of the first best only through objective bias, which lowers a^{SE} (Corollary 2) and suppresses the operational clarification rate.

6 Model II: A Screening (Menu) Policy

Now let the leader condition its ask rate on the user's declared specification level (equivalently, by the revelation principle, offer a menu of bundles that users self-select into). For tractability and clean analytical exposition, we restrict attention to two types $\kappa_L < \kappa_H$ with population shares f_L, f_H (the continuum-type extension using the standard Mirrlees technique is discussed in Section 12). Let the leader choose bundles $(m_L, a_L), (m_H, a_H)$ subject to

$$\text{IC}_\kappa: \;\; U(m_\kappa, a_\kappa; \kappa) \geq U(m_{\kappa'}, a_{\kappa'}; \kappa), \qquad \text{IR}_\kappa: \;\; U(m_\kappa, a_\kappa; \kappa) \geq \underline{U},$$

to maximize $f_L \Pi_{\kappa_L}(m_L, a_L) + f_H \Pi_{\kappa_H}(m_H, a_H)$. We normalize the follower's reservation utility to $\underline{U} = 0$, representing complete task abandonment without AI assistance (with V large enough that this constraint never binds). The economically relevant outside option is stronger. Because users can type any prompt, the assistant must also specify a response to off-menu specification levels, and the harshest admissible one is not to ask ($a = 0$). A user can therefore always secure $\underline{U}(\kappa) \equiv \max_{m \in [0,k]} U(m, 0; \kappa)$, which equals $V - \frac{\kappa}{2}k^2$ for $\kappa \leq \Lambda/k$ and $V - \Lambda k + \frac{\Lambda^2}{2\kappa}$ otherwise. Lemma 3 below shows that the optimal menu respects this endogenous outside option for both types, so it can be implemented with free prompt choice.

6.1 Single-crossing

The marginal rate of substitution between m and a along an indifference curve in (m, a)-space is given by the slope $\mathrm{MRS}(m, a; \kappa) \equiv -\frac{\partial U/\partial m}{\partial U/\partial a}$. From user utility $U(m, a; \kappa) = V - \Lambda(1 - a)(k - m) - \frac{\kappa}{2}m^2 - ac_Q(k - m)$, we have $\partial U/\partial m = \Lambda(1 - a) + ac_Q - \kappa m$ and $\partial U/\partial a = s(k - m)$, where $s \equiv \Lambda - c_Q > 0$ is the net social gain from asking. Hence,

$$\mathrm{MRS}(m, a; \kappa) \;=\; \frac{\kappa m - [\Lambda - as]}{s(k - m)}, \qquad \frac{\partial \mathrm{MRS}}{\partial \kappa} \;=\; \frac{m}{s(k - m)} \;>\; 0,$$

which holds strictly for all interior specification levels $0 < m < k$ under the maintained condition $s > 0$. Higher-κ types therefore have strictly steeper indifference curves in (m, a)-space: they require more compensating clarification service a per unit increase in specification effort m. This satisfies the standard Spence–Mirrlees single-crossing property (Spence, 1973) and underwrites the screening analysis below.

6.2 No distortion without bias

Proposition 6 (Efficiency survives private information when the leader is unbiased)**.** *If $\mu_A = 1$ and $\lambda_A = c_Q$ (so $\Pi_\kappa \equiv U(\cdot; \kappa)$ for every κ), then implementing each type's own first-best bundle, $(m_\kappa, a_\kappa) = (m^{FB}(\kappa), a^{FB}(\kappa))$ from Proposition 3, is incentive compatible, and no further mechanism design can do better for the leader.*

Proof. $(m^{FB}(\kappa), a^{FB}(\kappa))$ maximizes $U(\cdot; \kappa)$ by definition, so for any $\kappa' \neq \kappa$,

$$U(m^{FB}(\kappa), a^{FB}(\kappa); \kappa) \;\geq\; U(m^{FB}(\kappa'), a^{FB}(\kappa'); \kappa)$$

holds trivially: no bundle can give type κ more utility than its own maximizer. IC holds with no distortion needed. Since $\Pi_\kappa \equiv U(\cdot; \kappa)$, the leader's payoff from serving each true type is maximized at that type's own first best, so this menu is also leader-optimal. □

This result highlights a fundamental distinction between human–AI coordination and classical adverse selection (such as Baron–Myerson or Laffont–Tirole). In regulatory and auction models, the principal pays monetary transfers to extract surplus and thus treats informational rents as a pure loss. Here, by contrast, the leader's objective can coincide directly with user welfare; when it does, incentive compatibility is costless, and private information incurs no welfare penalty. Distortion in this setting therefore requires a wedge between the leader's objective and user utility, whether that wedge is bias or a real provider cost (Remark 3); asymmetric information alone causes none.

Remark 9 (Degeneracy of the Unbiased Menu on the Asking Margin)**.** *Under Proposition 3, $a^{FB}(\kappa) = 1$ for every type in the population on the physically feasible domain $m \leq k$. The unbiased*

first-best menu is therefore $(m^{FB}(\kappa), 1) = (\min\{k, c_Q/\kappa\}, 1)$*, which offers uniform asking* $a = 1$ *across all types and leaves followers to freely choose their preferred specification effort. This menu does not screen across the asking margin: scalar pooling at* $a^{SE} = 1$ *(Corollary 4) implements each type's first best identically. Consequently, under linear risk on the feasible domain, the welfare gap between pooling and screening at zero bias is identically zero (*$\Pi_{II} - \Pi_I = 0$*). A positive instrument-restriction gap at zero bias emerges only in settings where first best itself prescribes different asking policies across types. Under the exact conjunctive payoff this happens on long tasks, where terse users guess at their first best and diligent users ask (Section 12). Ordered attribute stakes produce it with the opposite sorting (Proposition 8).*

6.3 Distortion under bias

When $\Pi_\kappa \neq U(\cdot;\kappa)$ (bias present), the leader's preferred bundle per type,

$$(m^B_\kappa, a^B_\kappa) \;=\; \arg\max_{m,a} \Pi_\kappa(m, a)$$

(computable in closed form by the same method as Proposition 3, with Λ, c_Q replaced by their μ_A, λ_A-weighted counterparts), no longer automatically satisfies IC, because Π_κ and the user's true $U(\cdot;\kappa)$ now rank bundles differently.

Proposition 7 (Second-Best Screening Menu and Distortion Regimes)**.** *Assume* $\Lambda > c_Q$*, under-asking bias (*$0 < \lambda_A - c_Q < \mu_A(\Lambda - c_Q)$*, so that* $c_Q < c_Q^{eff} < \Lambda$*), and cost heterogeneity* $\kappa_L < c_Q^{eff}/k < \kappa_H$*. Let population shares be* $f_L, f_H > 0$ *with* $f_L + f_H = 1$*, define the relative share weight* $t \equiv \frac{f_H}{f_L}\frac{s-\tilde\rho}{s}$*, and define the stationary candidate:*

$$m^*_H \;\equiv\; \frac{\Lambda\tilde\rho}{s(\kappa_H - \kappa_L) + \tilde\rho\kappa_L}.$$

Define the leader's unconstrained bundle for type H*,* $m^B_H \equiv c_Q^{eff}/\kappa_H$ *(with* $a^B_H = 1$*), the complexity thresholds* $k_{slack} \equiv \frac{2c_Q}{\kappa_L} - m^B_H$ *and* $k_{crit} \equiv \frac{2c_Q}{\kappa_L} - m^*_H$ *(so* $k_{slack} < k_{crit}$ *if and only if* $m^*_H < m^B_H$*), and the saturation threshold* $\bar\kappa_L \equiv \frac{c_Q^{eff} + tc_Q}{k(1+t)}$*. Then the four regimes below partition the parameter space; in regimes 2–4, the closed forms describe the optimal menu whenever the saturation condition stated for that regime holds (otherwise item 5 applies):*

1. ***Unconstrained slack regime (*$k \le k_{slack}$*):*** *The unconstrained menu* $((k,0),(m^B_H,1))$ *satisfies* IC_L *(*$\kappa_L(k + m^B_H) \le 2c_Q$*). Both types receive their unconstrained preferred bundles, with zero distortion in effort and asking. This regime contains all* $\kappa_L \le c_Q/k$*.*

2. ***Effort-distortion regime (*$k_{slack} < k \le k_{crit}$***, non-empty only when** $m^*_H < m^B_H$***):*** *Under the saturation condition* $f_H\,(c_Q^{eff} - \kappa_H m^{SB}_H) \le f_L\,(c_Q^{eff} - \kappa_L k)$*,* IC_L *binds, type* L *receives* $(k,0)$*, and asking remains at the leader-preferred level* $a_H = 1$*. The distortion is purely a downward effort distortion on the high-cost type:*

$$m^{SB}_H \;=\; \frac{2c_Q}{\kappa_L} - k \;<\; m^B_H, \qquad a^{SB}_H \;=\; 1.$$

3. ***Asking-distortion regime (*$k > \max\{k_{slack}, k_{crit}\}$ ***and*** $k \ge m^*_H$***):*** *Under the saturation condition* $\kappa_L \le \bar\kappa_L$*, the leader sets* $(m^{SB}_L, a^{SB}_L) = (k, 0)$*, and the high-cost contract is unique and*

given in closed form by:

$$m_H^{SB} \;=\; m_H^*, \qquad a_H^{SB} \;=\; 1 \;-\; \frac{\frac{\kappa_L(k+m_H^*)}{2} - c_Q}{s} \;\in\; \Big(1 - \frac{\tilde{\rho}}{s},\, 1\Big).$$

The asking distortion is therefore bounded by the relative bias: $0 < 1 - a_H^{SB} < \tilde{\rho}/s$. It intensifies as κ_L increases toward $\bar{\kappa}_L$. At $m_L = k$, zero residual attributes remain ($k - m_L = 0$), so a_L is payoff-irrelevant to both parties.

4. ***Full-specification regime*** *($k_{\textbf{slack}} < k < m_H^*$):* *When cost heterogeneity is narrow, $m_H^* > k$ and the feasibility constraint $m_H \le k$ binds. Under the saturation condition $\kappa_L \le \bar{\kappa}_L$, both types specify fully, $m_L^{SB} = m_H^{SB} = k$; residual attributes vanish, asking is payoff-irrelevant, and the menu is payoff-equivalent to pooling at $(k, 0)$.*

5. ***Boundary relaxation:*** *When the saturation condition of regime 2, 3, or 4 fails (in regimes 3–4, when $\bar{\kappa}_L < \kappa_L < c_Q^{eff}/k$), the leader instead offers type L clarification service ($a_L^{SB} = 1$, $m_L^{SB} < k$). This raises U_L, relaxes IC_L, and weakly moderates the distortion on type H relative to the saturation formula without reversing its sign.*

In regimes 1–4, type L is held exactly to its outside option, $U_L = \underline{U}(\kappa_L)$, while type H earns a strictly positive rent over $\underline{U}(\kappa_H)$ in regimes 1–3 (except at the knife-edge $k = m_H^*$).

Lemma 3 (Implementability with free prompt choice)**.** *Suppose the assistant answers every off-menu specification level with $a = 0$. Then, under the hypotheses of Proposition 7, no type gains by choosing an off-menu prompt: each type's menu bundle yields at least $\underline{U}(\kappa) = \max_{m \in [0,k]} U(m, 0; \kappa)$.*

Proof. For type L, $\kappa_L k < c_Q^{eff} < \Lambda$, so $U(\cdot, 0; \kappa_L)$ is increasing on $[0, k]$ and $\underline{U}(\kappa_L) = U(k, 0; \kappa_L)$; type L's bundle yields at least this (with equality at $(k, 0)$, and strictly more under boundary relaxation). For type H, let $m^o \equiv \min\{k, \Lambda/\kappa_H\}$ be its unassisted optimum. In every regime the assigned effort satisfies $m_H^{SB} \le m^o$: $m_H^B = c_Q^{eff}/\kappa_H < \Lambda/\kappa_H$, the effort-regime level lies below m_H^B, and $m_H^* < \Lambda/\kappa_H$ because $(s - \tilde{\rho})(\kappa_H - \kappa_L) > 0$ (in regime 4, $k < m_H^* < \Lambda/\kappa_H$, so $m^o = k = m_H^{SB}$).

Case 1: IC_L binds (regimes 2–5). Then $U(m_H^{SB}, a_H^{SB}; \kappa_L) = U_L \ge \underline{U}(\kappa_L) \ge U(m^o, 0; \kappa_L)$. Using $U(m, a; \kappa_H) = U(m, a; \kappa_L) - \frac{\kappa_H - \kappa_L}{2} m^2$,

$$U(m_H^{SB}, a_H^{SB}; \kappa_H) - U(m^o, 0; \kappa_H) \;\ge\; \frac{\kappa_H - \kappa_L}{2}\Big[(m^o)^2 - (m_H^{SB})^2\Big] \;\ge\; 0.$$

Case 2: slack regime, bundle $(m_H^B, 1)$. If $\kappa_H \le \Lambda/k$, then $m^o = k$ and $U(m_H^B, 1; \kappa_H) - U(k, 0; \kappa_H) = (k - m_H^B)\big[\frac{\kappa_H k + c_Q^{eff}}{2} - c_Q\big] > 0$. If $\kappa_H > \Lambda/k$, then $m^o = \Lambda/\kappa_H$, and direct substitution gives

$$U(m_H^B, 1; \kappa_H) - U(m^o, 0; \kappa_H) \;=\; sk - \frac{s(\Lambda + c_Q) + \tilde{\rho}^2}{2\kappa_H} \;>\; sk - \frac{2s\Lambda}{2\kappa_H} \;=\; s\Big(k - \frac{\Lambda}{\kappa_H}\Big) \;>\; 0,$$

using $\tilde{\rho} < s$. □

The schedule in Lemma 3 is discontinuous: a prompt slightly longer or shorter than m_H^{SB} receives no clarification at all. The discontinuity is not needed.

Lemma 4 (Continuous implementation)**.** *Take any menu $\{(m_L, a_L), (m_H, a_H)\}$ that satisfies IC_L and IC_H and gives each type at least its outside option, $U_\theta \equiv U(m_\theta, a_\theta; \kappa_\theta) \ge \underline{U}(\kappa_\theta)$; by Lemma 3 this covers the menus of Proposition 7. For $m < k$ let*

$$\bar{a}_\theta(m) \;\equiv\; \frac{U_\theta - V + \Lambda(k - m) + \frac{\kappa_\theta}{2} m^2}{s\,(k - m)}$$

be the ask rate at which type θ is indifferent between prompt m and its own bundle, and let $\alpha(m) \equiv \min\{1, \bar{a}_L(m), \bar{a}_H(m)\}$, extended to $m = k$ by its limit.

(i) α is continuous on $[0, k]$, takes values in $[0, 1]$, and $\alpha(m_\theta) = a_\theta$ for each type with $m_\theta < k$.

(ii) Under the schedule $a(m) = \alpha(m)$, no type gains from any prompt $m \in [0, k]$, so the menu is implemented with free prompt choice (with ties resolved in the leader's favour, as is standard).

(iii) α is the most generous such schedule: any schedule under which neither type gains from deviating satisfies $a(m) \leq \alpha(m)$ for all $m < k$.

(iv) In regimes 2 and 3 of Proposition 7, $\alpha(m) = \bar{a}_L(m) = [\Lambda - \kappa_L(k + m)/2]/s$ for $m \geq m_H^{SB}$: above the terse type's level, the assistant asks less per open attribute the more the user specifies.

(v) A minimum-effort threshold, asking at rate a_H at every $m \geq m_H$ and never below, does not implement the menu in regimes 2 and 3 of Proposition 7, nor in Corollary 6.

Proof. For $m < k$, U is affine and strictly increasing in a with slope $s(k - m) > 0$, so $\bar{a}_\theta(m)$ is well defined and continuous on $[0, k)$. Since $U_\theta \geq \underline{U}(\kappa_\theta) \geq U(m, 0; \kappa_\theta)$, $\bar{a}_\theta(m) \geq 0$. At m_θ, $\bar{a}_\theta(m_\theta) = a_\theta$, and the other type's incentive constraint, $U(m_\theta, a_\theta; \kappa_{\theta'}) \leq U_{\theta'}$, gives $\bar{a}_{\theta'}(m_\theta) \geq a_\theta$; with $a_\theta \leq 1$ this yields $\alpha(m_\theta) = a_\theta$. As $m \to k$ the numerator of $\bar{a}_\theta$ tends to $U_\theta - U(k, 0; \kappa_\theta) \geq 0$; if it is positive, $\bar{a}_\theta \to +\infty$, and if it is zero, $U(k, 0; \kappa_\theta) = \underline{U}(\kappa_\theta)$ requires $\Lambda \geq \kappa_\theta k$ and l'Hôpital's rule gives the limit $(\Lambda - \kappa_\theta k)/s \geq 0$. Hence α has a limit in $[0, 1]$ at k, proving (i). For (ii), $\alpha(m) \leq \bar{a}_\theta(m)$ and monotonicity in a give $U(m, \alpha(m); \kappa_\theta) \leq U_\theta$ for $m < k$, and $U(k, \cdot\,; \kappa_\theta) = U(k, 0; \kappa_\theta) \leq U_\theta$. For (iii), if $a(m) > \alpha(m)$ at some $m < k$, then $a(m) > \bar{a}_\theta(m)$ for some type, which then strictly gains at m. For (iv), type L receives $(k, 0)$ at its outside option, so $U_L = V - \frac{\kappa_L}{2}k^2$, and substitution gives the formula for $\bar{a}_L$. Moreover $\bar{a}_H(m) - \bar{a}_L(m) = [U_H - U_L + \frac{\kappa_H - \kappa_L}{2}m^2]/[s(k - m)]$ has a numerator that is increasing in m and, because IC_L binds, vanishes at m_H^{SB}; so $\bar{a}_H \geq \bar{a}_L$ above m_H^{SB}, and $\bar{a}_L \leq a_H^{SB} \leq 1$ there. For (v), in regimes 2 and 3 type L's payoff at rate a_H^{SB} is strictly concave in m and equals U_L both at m_H^{SB} (binding IC_L) and at k (where asking is irrelevant), so it strictly exceeds U_L at every $m \in (m_H^{SB}, k)$. In Corollary 6, type L's payoff at $a = 1$ peaks at c_Q/κ_L, which lies strictly between $m_H = c_Q^{eff}/\kappa_H$ and $m_L = c_Q^{eff}/\kappa_L$ under that result's bound on $\tilde{\rho}$, so type L would shorten its prompt to c_Q/κ_L and keep full clarification. □

Lemma 4 removes the discontinuity but not the non-monotonicity. In regimes 2 and 3, every implementing schedule must ask less about each open attribute on prompts somewhat longer than m_H^{SB}, by (iii) and (iv), because otherwise the diligent type would trade part of its specification for clarification. This is the screening cost in continuous form. Upward deviations need no penalty only when no type gains from them: in Corollary 6, both types' payoffs at $a = 1$ fall beyond m_L, so $\alpha(m) = 1$ for every $m \geq m_L$. A numerical audit of 212 menus confirms (i)–(v); under a minimum-effort threshold, type L gains in all 28 sampled menus with binding IC_L and in all 12 Corollary 6 menus.

The proof of Proposition 7 is given in Appendix A.

Remark 10 (Active Constraints and Effort versus Asking Distortions)**.** *In classical transfer-based screening (such as Baron–Myerson), the efficient type's incentive constraint (IC_L in our labeling) and the inefficient type's participation constraint (IR_H) bind: the efficient type earns an information rent and the inefficient type is distorted and held to its outside option. Here, in the saturation regimes, IC_L and IR_L bind (against the endogenous outside option) while IR_H is slack: the efficient specifier earns no rent, and the distorted high-cost type keeps a rent. Without transfers, the leader cannot reward type L for not mimicking, because asking service is worthless to a user who specifies*

everything; it can only make the mimicked bundle less attractive. Solving the full four-variable constrained problem reveals five distinct operational regimes:

1. Unconstrained slack regime ($k \le k_{\text{slack}}$): *When task complexity is small, the temptation of type L to mimic is absent ($\kappa_L(k+m_H^B) \le 2c_Q$), so IC_L and IC_H are strictly slack at the unconstrained menu.*

2. Effort-distortion regime ($k_{\text{slack}} < k \le k_{\text{crit}}$): *Type L faces a temptation to mimic, but the assistant absorbs the entire distortion through specification effort ($m_H = \frac{2c_Q}{\kappa_L} - k < m_H^B$) while preserving full clarification service ($a_H = 1$). Here IC_L binds while asking is undistorted.*

3. Adverse-selection asking distortion regime ($k > k_{\text{crit}}$ with $k \ge m_H^*$): *Specification distortion alone is insufficient, forcing the assistant to distort asking downward ($a_H^{SB} < 1$) alongside effort $m_H = m_H^*$.*

4. Full-specification regime ($k_{\text{slack}} < k < m_H^*$): *Narrow cost heterogeneity forces the interior effort candidate above task complexity ($m_H^* > k$), causing the physical ceiling $m_H \le k$ to bind; both types specify fully and the menu is payoff-equivalent to pooling at $(k, 0)$ (unless boundary relaxation applies).*

5. Prohibitive friction regime ($c_Q^{eff} \ge \Lambda$, equivalently $\lambda_A - c_Q \ge \mu_A(\Lambda - c_Q)$): *Asking friction exceeds the entire task risk resolved, so the leader prefers zero asking for all types. Clarification is uniformly shut down ($a_L = a_H = 0$) and each type stays at its unassisted optimum $\min\{k, \Lambda/\kappa\}$; when both types saturate at k, they receive the identical bundle $(k, 0)$ and IC_L and IC_H hold with equality trivially. This co-activity reflects technological shutdown, not an adverse-selection screening distortion.*

Remark 11 (Directionality of Policy Misalignment)**.** *The downward distortion in Proposition 7 is specific to the under-asking bias case ($0 < \lambda_A - c_Q < \mu_A(\Lambda - c_Q)$), which reflects the empirical reality of turn-penalizing satisfaction metrics. Under an over-asking bias ($\lambda_A < c_Q$), the direction of distortion can reverse, as the leader would favor excessive clarification.*

7 Comparing the Two Regimes

Corollary 5 (Menus weakly dominate pooling)**.** *The leader's optimal expected payoff under Model II is weakly greater than under Model I: any pooling policy a is replicated by the menu that offers each type its own best response at ask rate a, $\{(m^*(\kappa_L; a), a), (m^*(\kappa_H; a), a)\}$. This menu is incentive compatible, because each type's bundle maximizes its utility among all bundles with ask rate a, so the optimal menu is at least as good.*

Figure 5 shows the leader-payoff gap $\Pi_{II} - \Pi_I$ across bias and type heterogeneity, and Figure 4 shows the corresponding distortion of the high-cost type's ask rate. User welfare need not follow the leader's ranking. Whenever pooling asks universally, users attain their first best, so pooling weakly beats any screening menu in user welfare; the next result gives the gap exactly.

Corollary 6 (Effort dictation: the pooling advantage under mild bias is second order)**.** *Suppose $c_Q < c_Q^{eff} < \Lambda$, both types satisfy $\kappa \ge c_Q^{eff}/k$, and $\tilde{\rho} \le c_Q(\kappa_H - \kappa_L)/(\kappa_H + \kappa_L)$. Then the optimal screening menu gives each type the leader's preferred bundle $(c_Q^{eff}/\kappa, 1)$, it is implementable with off-menu prompts answered by $a = 0$, and user welfare under screening is*

$$W_{screen} \;=\; W^{FB} \;-\; \frac{\tilde{\rho}^2}{2}\,\mathbb{E}\Big[\frac{1}{\kappa}\Big].$$

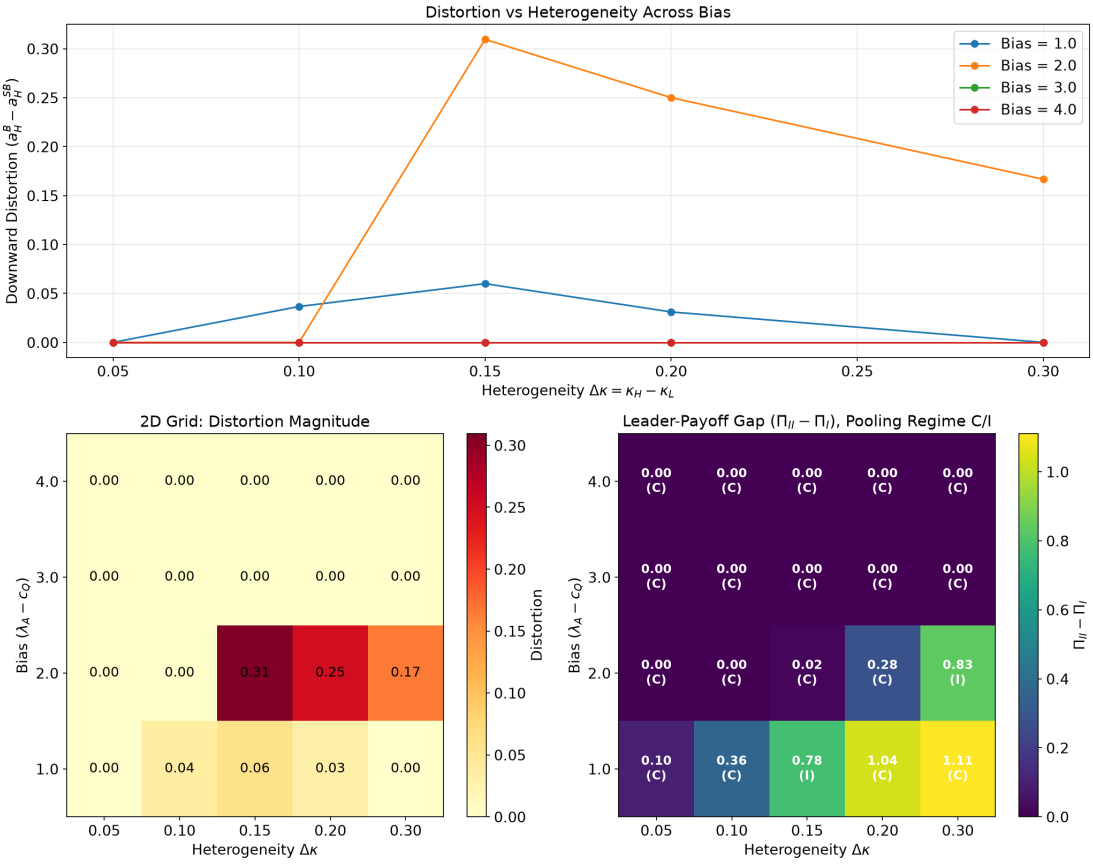


Figure 4: Screening across cost heterogeneity $\Delta\kappa = \kappa_H - \kappa_L$ ($\kappa_L = 0.3$) and bias $\lambda_A - c_Q$. Top: downward asking distortion $a_H^B - a_H^{SB}$. Bottom left: the same distortion as a grid. Bottom right: the leader-payoff gap $\Pi_{II} - \Pi_I$, with the pooling regime marked C (corner) or I (interior).

If, in addition, the pooling leader selects $a^{SE} = 1$*, then* $W_{pool} = W^{FB}$ *and* $W_{pool} - W_{screen} = \frac{\tilde{\rho}^2}{2}\mathbb{E}[1/\kappa] > 0$.

Proof. For each type, $\Pi_\kappa = \mu_A[V - \Lambda(1-a)(k-m) - \frac{\kappa}{2}m^2 - c_Q^{eff}a(k-m)]$ has slope $\mu_A(\Lambda - c_Q^{eff})(k-m) \geq 0$ in a, so the leader's preferred bundle is $a = 1$ and $m = \min\{k, c_Q^{eff}/\kappa\} = c_Q^{eff}/\kappa$. These bundles maximize each Π_κ pointwise, so they are optimal if they are incentive compatible. At a common $a = 1$, $U(m, 1; \kappa) - U(m', 1; \kappa) = (m - m')[c_Q - \kappa(m + m')/2]$. With $m_L > m_H$, IC_L holds iff $\kappa_L(m_L + m_H) \leq 2c_Q$, i.e. $c_Q^{eff}(1 + \kappa_L/\kappa_H) \leq 2c_Q$, which is the stated bound on $\tilde{\rho}$; IC_H holds because $\kappa_H(m_L + m_H) = c_Q^{eff}(\kappa_H/\kappa_L + 1) > 2c_Q$. Implementability follows from the computation in Case 2 of the proof of Lemma 3, which applies to any type whose bundle is $(c_Q^{eff}/\kappa, 1)$ with $c_Q^{eff}/\kappa \leq k$. Finally, $U(\cdot, 1; \kappa)$ is quadratic with curvature κ and peak at $m^{FB} = c_Q/\kappa < k$, so $U(m^{FB}, 1; \kappa) - U(c_Q^{eff}/\kappa, 1; \kappa) = \frac{\kappa}{2}(\tilde{\rho}/\kappa)^2 = \tilde{\rho}^2/(2\kappa)$. Averaging over types gives W_{screen}. Under pooling at $a = 1$ each type chooses m^{FB}, so $W_{\text{pool}} = W^{FB}$. □

Beyond payoff dominance, the central comparison lies in the structural decomposition of welfare losses across the two regimes, enabled by Propositions 4–7:

- **Model I (pooling):** Under the linear-risk specification on the physically feasible domain $m \in [0, k]$, an unbiased assistant selects $a^{SE} = 1$, which implements the first-best policy $a^{FB}(\kappa) = 1$ for every type in the population (Proposition 3). Consequently, in the absence of bias ($\lambda_A = c_Q$), the pooling restriction incurs zero welfare loss ($\Pi_{II} - \Pi_I = 0$). Welfare loss under pooling arises when objective bias is present ($\lambda_A > c_Q$), which suppresses the single ask rate a^{SE} below the socially efficient level (Corollary 2), or in the exact conjunctive model on long tasks (Section 12), where the terse type's first best is to guess while the diligent type asks, and a uniform rate cannot serve both.
- **Model II (screening)** incurs welfare loss from bias *alone* (Proposition 6): in the absence of bias, screening recovers full efficiency despite private cost information. Under bias, the loss takes the form of the downward distortion characterized in Proposition 7. This represents a fundamentally different mechanism from the instrument-restriction loss of pooling: it stems not from an inability to condition on observed effort, but from the requirement to prevent opportunistic mimicking across types.

7.1 Heterogeneous attribute stakes and the zero-bias benchmark

The zero-bias degeneracy of Remark 9 comes from attribute symmetry: under linear risk every open attribute is worth the same net gain s from a question, so asking is first best for every type. Let attributes differ in stakes instead. Index them by $x \in [0, k]$ in decreasing order of stake $\Lambda(x) = L(x)(1 - g)$, with $\Lambda(\cdot)$ continuous and strictly decreasing, and let the assistant ask at a uniform rate a about every open attribute. Such an attribute-blind policy is natural when the assistant sees which requirements are missing but not how much each matters to this user. For $a < 1$ the residual cost of an open attribute, $(1 - a)\Lambda(x) + ac_Q$, is decreasing in x, so a user who specifies m attributes specifies the m most important ones (at $a = 1$ the user is indifferent, and we keep this order). Then

$$U(m, a; \kappa) \;=\; V - \int_m^k \big[(1 - a)\Lambda(x) + ac_Q\big]\, dx - \frac{\kappa}{2}m^2,$$

which reduces to the baseline when $\Lambda(x) \equiv \Lambda$. Let $\bar{\Lambda} \equiv \frac{1}{k}\int_0^k \Lambda(x)\,dx$ and $G(\kappa) \equiv \max_m U(m,1;\kappa) - \max_m U(m,0;\kappa)$.

Proposition 8 (Heterogeneous stakes: a first-best asking threshold)**.** *Suppose $\Lambda(k) < c_Q < \bar{\Lambda}$, and let $x^* \in (0,k)$ solve $\Lambda(x^*) = c_Q$. There is a unique $\hat{\kappa} > c_Q/x^*$ such that the first-best uniform ask rate is*

$$a^{FB}(\kappa) = 0 \quad \text{for } \Lambda(k)/k < \kappa < \hat{\kappa}, \qquad a^{FB}(\kappa) = 1 \quad \text{for } \kappa > \hat{\kappa},$$

and both rates are optimal for $\kappa \leq \Lambda(k)/k$, where every user specifies fully. Diligent users guess and terse users ask. Consequently:

(i) In any two-type population with $\kappa_L \in (\Lambda(k)/k, \hat{\kappa})$ and $\kappa_H > \hat{\kappa}$, an unbiased pooling leader falls strictly short of the first best, while unbiased screening attains it (the argument of Proposition 6 does not use the functional form). The zero-bias instrument-restriction gap is strictly positive.

(ii) If the assistant can instead condition on which attributes are open, asking about attribute x if and only if $\Lambda(x) > c_Q$ is first best for every type, and pooling and screening coincide again at zero bias.

Proof. For each a, $U(\cdot, a; \kappa)$ is strictly concave in m ($U_{mm} = (1-a)\Lambda'(m) - \kappa < 0$), with unique maximizer $m_a(\kappa)$, so the envelope theorem gives $G'(\kappa) = [m_0(\kappa)^2 - m_1(\kappa)^2]/2$. Here $m_1 = \min\{k, c_Q/\kappa\}$, and $m_0 = k$ if $\Lambda(k) \geq \kappa k$ and otherwise solves $\Lambda(m_0) = \kappa m_0$; the function $\Lambda(m) - \kappa m$ is strictly decreasing. If $\kappa > c_Q/x^*$, then $m_1 = c_Q/\kappa < x^*$, so $\Lambda(m_1) > c_Q = \kappa m_1$ and $m_0 > m_1$: $G' > 0$. If $\Lambda(k)/k < \kappa < c_Q/x^*$, then either $m_1 = c_Q/\kappa \in (x^*, k)$, where $\Lambda(m_1) < \kappa m_1$, or $m_1 = k > m_0$; in both cases $m_0 < m_1$ and $G' < 0$. If $\kappa \leq \Lambda(k)/k < c_Q/k$, then $m_0 = m_1 = k$ and $G = 0$. So G is zero up to $\Lambda(k)/k$, strictly decreasing up to c_Q/x^*, and strictly increasing beyond. As $\kappa \to \infty$, $m_0, m_1 \leq \Lambda(0)/\kappa \to 0$, so $G(\kappa) \to \int_0^k [\Lambda(x) - c_Q]\,dx = k(\bar{\Lambda} - c_Q) > 0$. By continuity G has a unique zero $\hat{\kappa} > c_Q/x^*$, is negative on $(\Lambda(k)/k, \hat{\kappa})$, and is positive beyond. Because U is affine in a at each m, $\max_m U(m, a; \kappa)$ is convex in a and is maximized over $[0,1]$ at $a \in \{0,1\}$, so $a^{FB}(\kappa) = \mathbf{1}\{G(\kappa) > 0\}$. For (i), pooling payoffs are also convex in a, so the pooling optimum is $a = 0$ or $a = 1$, and each falls short of the first best by $f_H G(\kappa_H) > 0$ or $f_L|G(\kappa_L)| > 0$. For (ii), the targeted policy charges each open attribute $\min\{\Lambda(x), c_Q\}$, the least cost available, whatever the type and whatever was specified, so it weakly dominates every policy for every type and does not depend on κ. □

The sorting is the opposite of the exact conjunctive model, where terse users guess (Section 12). In both cases the sign of G' follows the envelope identity; what differs is whether the attributes a user leaves open become more or less pivotal as the prompt lengthens. Conjunctive complementarity makes the last open attributes the most valuable to resolve, while ordered stakes make them the least valuable. The gap is small in magnitude. With $k = 10$, $c_Q = 2$, and $\Lambda(x) = 9 - 0.8x$ (so $\bar{\Lambda} = 5$, matching the benchmark, and $\Lambda(k) = 1$), $\hat{\kappa} \approx 0.31$; for $\kappa_L = 0.3$, $\kappa_H = 0.5$ and equal shares, the zero-bias gap is 0.08 on $V = 100$, while targeted asking adds a further 0.55 over the uniform first best. In this example, what matters most for welfare is whether the assistant can target its questions, not how users sort across types. Screening under bias with heterogeneous stakes is left open.

Remark 12 (Welfare vs Leader Payoff Ordering Across Bias Bands)**.** *While screening weakly dominates pooling in leader payoff ($\Pi_{II} \geq \Pi_I$) by revealed preference (Corollary 5), the ranking of user welfare $W \equiv \mathbb{E}[U]$ depends non-monotonically on provider asking friction λ_A (evaluated at $\kappa_L = 0.3, \kappa_H = 0.5, f_L = 0.5, k = 10, L = 10, g = 0.5, c_Q = 2, V = 100$, and $\mu_A = 1.0$):*

1. Mild bias ($c_Q < \lambda_A < 3.0$): *Pooling asks universally ($a^{SE} = 1$), so users attain first-best welfare, $W_{pool} = W^{FB} = 85.33$, and pooling weakly beats any screening menu. For $\lambda_A \leq 2.5$ the hypotheses of Corollary 6 hold ($\kappa_L = 0.3 \geq c_Q^{eff}/k$ and $\tilde{\rho} \leq c_Q(\kappa_H - \kappa_L)/(\kappa_H + \kappa_L) = 0.5$). The loss under screening then arises entirely through* effort dictation*: asking is undistorted ($a_L = a_H = 1$), but the biased assistant demands more upstream specification than users would choose ($m_L = 8.33 > 6.67$ and $m_H = 5.00 > 4.00$ at $\lambda_A = 2.5$) to economize on its own query costs. The gap is exactly $\tilde{\rho}^2 \mathbb{E}[1/\kappa]/2$, which is 0.33 at $\lambda_A = 2.5$, or 0.4% of W.*

2. Knife-edge ($\lambda_A = 3.0$): *The pooling leader is exactly indifferent between $a = 1$ and $a = 2/3$ ($\Pi = 80.67$ at both), so W_{pool} equals 85.33 or 82.00 depending on the tie-break. Screening gives $(m_L, a_L) = (8.67, 1)$ and $(m_H, a_H) = (5.56, 0.97)$, with $W_{screen} = 84.52$; here IC_L binds and effort dictation is accompanied by a small downward asking distortion. The sign of the welfare gap at this point is not determined by the model.*

3. Intermediate bias ($3.0 < \lambda_A < 3.5$): *Pooling drops to $a^{SE} = 2/3$, the kink at which type L's best response leaves the boundary $m = k$, and user welfare falls to $W_{pool} = 82.00$, while screening still delivers $W_{screen} \in (83.0, 83.6)$ (gap +1.57 at $\lambda_A = 3.25$). The welfare ranking reverses at $\lambda_A = 3.0$, where pooling stops asking universally.*

4. Moderate bias ($3.5 \leq \lambda_A < 5.0$): *Pooling collapses to zero asking ($a^{SE} = 0$, yielding baseline unassisted welfare $W_{pool} = 80.00$), whereas screening continues to provide partial clarification to terse users ($a_H^{SB} \in [0.70, 0.81]$), delivering strictly higher user welfare ($W_{screen} = 82.45$ at $\lambda_A = 3.5$, gap +2.45; 81.53 at $\lambda_A = 4.0$, gap +1.53; 80.71 at $\lambda_A = 4.5$, gap +0.71).*

5. Prohibitive friction ($\lambda_A \geq 5.0$, i.e., $c_Q^{eff} \geq \Lambda$): *Clarification is priced out under both regimes, so user welfare ties at $W_{pool} = W_{screen} = 80.00$ (gap 0.00).*

The pooling breakpoints $\lambda_A = 3.0$ and 3.5 are exact. For $a \leq 2/3$, type L is clipped at $m = k$ and type H's contribution is $3(3 - 2b)a^2$ plus a constant, which is convex for $b < 1.5$ and flat at $b = 1.5$ ($\lambda_A = 3.5$). The two sides of the ranking differ in order of magnitude. Under mild bias the pooling advantage is quadratic in $\tilde{\rho}$ (Corollary 6). Once pooling withdraws clarification, the screening advantage reflects the asking surplus that pooling forgoes on the terse type's residual attributes, here up to 2.45. The ordering is computed for one calibration, and we do not claim that its breakpoints generalize. Under the exact conjunctive payoff, pooling can also lose to screening at zero bias, but only on long tasks (Section 12).

8 A Continuum of Users: The Optimal Clarification Standard

The two-type menu of Section 6 leaves open what a screening assistant should do when users differ continuously, which is the relevant case for a deployed policy. This section solves that problem. The answer is simpler than the two-type menu, and in one respect it reverses it. Under moderate bias the optimal policy is a *minimum-specification standard*: the assistant asks about every open attribute once the prompt states at least $\bar{m}$ requirements. Diligent users are unaffected, terse users pool at the standard, and no attribute is left to a guess.

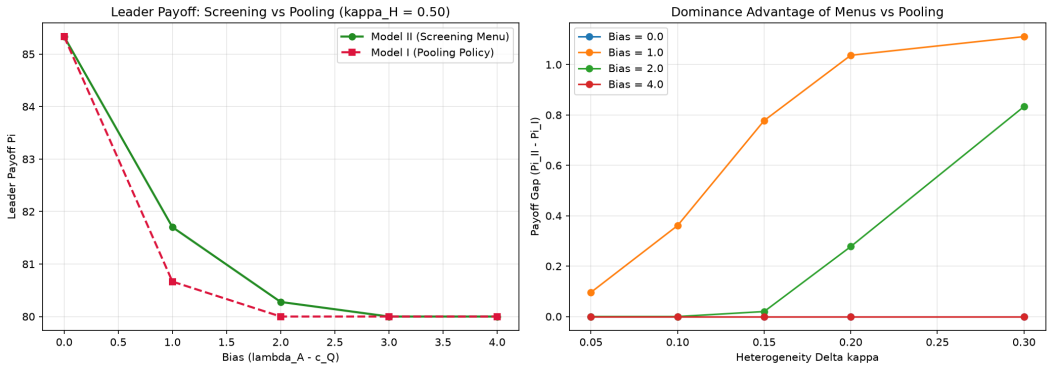


Figure 5: Leader payoff under Model II (screening menu) versus Model I (pooling policy) across assistant bias $\lambda_A - c_Q$ and type heterogeneity $\kappa_H - \kappa_L$, illustrating weak dominance ($\Pi_{II} \geq \Pi_I$) and the zero-gap baseline at zero bias under linear risk.

8.1 Screening with a capped in-kind transfer

Let κ have a density $f > 0$ on $[\underline{\kappa}, \bar{\kappa}]$ with distribution function F, and let the leader offer a direct mechanism $(m(\kappa), a(\kappa))$, or equivalently a schedule $a(m)$ from which users choose. Write

$$U_0(m,\kappa) \;\equiv\; V - \Lambda(k-m) - \frac{\kappa}{2}m^2, \qquad x \;\equiv\; a\,s\,(k-m), \qquad U_1(m,\kappa) \;\equiv\; U_0(m,\kappa) + s(k-m),$$

so that U_0 is the payoff without clarification, U_1 the payoff with full clarification, and $U = U_0(m,\kappa) + x$ with $0 \leq x \leq s(k-m)$. By Lemma 2 the leader maximizes $\mathbb{E}[U - \tilde{\rho}\,a(k-m)] = \mathbb{E}[U_0(m,\kappa) + (1-r)\,x]$, where

$$r \;\equiv\; \tilde{\rho}/s \;\in\; (0,1)$$

under the maintained bias $c_Q < c_Q^{eff} < \Lambda$. This is quasilinear screening with an unusual transfer. The transfer x is paid in kind, the leader values paying it (weight $1 - r > 0$), and it is capped by the number of open attributes. The classical motive to extract information rents is therefore absent. What limits the leader is the cap $a \leq 1$.

Lemma 5 (Reformulation)**.** *A mechanism is incentive compatible if and only if $m(\cdot)$ is nonincreasing and $U(\kappa) = U(\bar{\kappa}) + \int_\kappa^{\bar{\kappa}} \frac{m(t)^2}{2}\,dt$. For every incentive-compatible mechanism, the leader's payoff (divided by μ_A) satisfies*

$$\Pi \;\leq\; J[m] \;\equiv\; \int_{\underline{\kappa}}^{\bar{\kappa}} \left[r\,U_0(m(\kappa),\kappa) f(\kappa) + (1-r)\,F(\kappa)\,\frac{m(\kappa)^2}{2} \right] d\kappa \;+\; (1-r)\min_\kappa G_m(\kappa),$$

where $G_m(\kappa) \equiv U_1(m(\kappa),\kappa) - \int_\kappa^{\bar{\kappa}} \frac{m(t)^2}{2}\,dt$, with equality when $U(\bar{\kappa}) = \min_\kappa G_m(\kappa)$ and $x \geq 0$ for every type.

Proof. Utility is quasilinear in x and $\partial^2 U_0/\partial m\,\partial\kappa = -m \leq 0$, so incentive compatibility is equivalent to monotonicity of m together with the envelope formula (Mirrlees, 1971; Myerson, 1981; Milgrom and Segal, 2002). By Fubini's theorem, $\mathbb{E}[U] = U(\bar{\kappa}) + \int F(\kappa)\,m(\kappa)^2/2\,d\kappa$, and the leader's payoff is $r\,\mathbb{E}[U_0] + (1-r)\,\mathbb{E}[U]$. The cap $a(\kappa) \leq 1$ reads $U(\kappa) \leq U_1(m(\kappa),\kappa)$, which by the envelope formula is $U(\bar{\kappa}) \leq G_m(\kappa)$. Hence $U(\bar{\kappa}) \leq \min G_m$, with equality the best the leader can do. □

The first consequence concerns the effort dictation of Corollary 6.

Proposition 9 (No effort dictation without pooling)**.** *In any incentive-compatible mechanism, if $a = 1$ on an interval I of types, then for almost every $\kappa \in I$, either $m'(\kappa) = 0$ or $m(\kappa) = c_Q/\kappa$.*

Proof. On I, $U(\kappa) = U_1(m(\kappa), \kappa)$. A monotone m is differentiable almost everywhere, and at such points $U'(\kappa) = (c_Q - \kappa m)\, m' - m^2/2$. Incentive compatibility gives $U' = -m^2/2$ almost everywhere, so $(c_Q - \kappa m)\, m' = 0$. □

In the two-type model the leader can demand more specification than either type would choose (Corollary 6), because the gap between the types leaves room to do so. With a continuum, an assistant that asks about every open attribute cannot make any user write more than she wants, except by pooling users on a common prompt length or by withholding clarification. The rest of the section shows that, under moderate bias, the leader pools.

8.2 The minimum-specification standard

For a threshold $\bar{m}$, the *standard* is the schedule $a(m) = 1$ if $m \geq \bar{m}$ and $a(m) = 0$ otherwise. Facing it, a user either meets the standard and receives full clarification, writing $\max\{\bar{m}, c_Q/\kappa\}$, or writes an unassisted prompt. If every type meets it, the leader's payoff is

$$\Pi_{\text{std}}(\bar{m}) \;=\; \mathbb{E}\Big[U_1(m_{\bar{m}}(\kappa), \kappa) - \tilde{\rho}\,\big(k - m_{\bar{m}}(\kappa)\big)\Big], \qquad m_{\bar{m}}(\kappa) \equiv \max\{\bar{m}, c_Q/\kappa\}.$$

Let $\bar{m}^*$ maximize Π_{std} on $[0, k]$ and let $\kappa_b \equiv \max\{\underline{\kappa}, c_Q/\bar{m}^*\}$ be the lowest type that pools at the standard. Define the *tail condition*

$$T(\tau) \;\equiv\; \int_{\tau}^{\bar{\kappa}} \Big[r f(t)\,(\Lambda - t\,\bar{m}^*) - (1-r)\big(1 - F(t)\big)\,\bar{m}^*\Big] dt \;\geq\; 0 \qquad \text{for all } \tau \in [\kappa_b, \bar{\kappa}].$$

Theorem 1 (Optimal clarification standard)**.** *Suppose $\underline{\kappa} \geq c_Q/k$, that $t \mapsto t f(t)$ is nondecreasing, and that $0 < \bar{m}^* < k$.*

(i) The standard $\bar{m}^*$ satisfies $\bar{m}^* = c_Q^{eff}/\mathbb{E}[\kappa \mid \kappa \geq \kappa_b]$, and it is nondecreasing in $\tilde{\rho}$.

(ii) If the tail condition holds, then under the standard $\bar{m}^*$ users with $\kappa < c_Q/\bar{m}^*$ write their first-best prompt c_Q/κ, users with $\kappa \geq c_Q/\bar{m}^*$ write exactly $\bar{m}^*$, every user receives full clarification, and the standard is optimal among all incentive-compatible mechanisms.

(iii) If the tail condition fails, the standard is not optimal: some mechanism that gives the tersest users partial clarification does strictly better.

(iv) The tail condition implies $\bar{m}^* \leq \Lambda/\bar{\kappa}$. If κ is uniform, the tail condition is equivalent to $\bar{m}^* \leq \Lambda/\bar{\kappa}$.

The proof is in Appendix B. It shows that the leader's value is bounded by a Lagrangian in which the minimum in Lemma 5 is replaced by an average under a suitable measure μ^*, and that the standard maximizes this Lagrangian over all nonincreasing effort schedules by an ironing argument. Part (i) has a direct reading: the standard is the leader's preferred prompt, c_Q^{eff}/κ, evaluated at the average type in the pool. Part (iv) says the standard may not exceed the prompt the tersest user would write with no help at all.

Three features distinguish this optimum from the two-type menu. First, the cost of bias falls entirely on effort, not on clarification: under moderate bias no user is asked less than every open

attribute, so the equilibrium residual $(1-a)(k-m)$ is zero, while terse users write longer prompts than they would choose. Second, the distortion falls on the terse users only; diligent users keep their first best, in contrast with Corollary 6, where both types are made to over-specify. Third, the optimum is implemented by a schedule that is monotone in prompt length, so the non-monotonicity of Lemma 4 is a feature of the two-type saturation regimes rather than of screening as such. Asking distortion of the kind in Proposition 7 reappears only when bias is large enough that the tail condition fails. As in Lemma 4, the step at $\bar{m}^*$ is not needed: the schedule $\alpha(m) = \min_\kappa \bar{a}_\kappa(m)$, where $\bar{a}_\kappa(m)$ is the ask rate that leaves type κ indifferent between m and its own prompt, is continuous because $\bar{a}_\kappa(m)$ is jointly continuous in (κ, m). It equals one at $\bar{m}^*$, where the pooled types are indifferent, and it implements the same allocation, so clarification can rise smoothly to one as a prompt approaches the standard.

The welfare cost of bias is also smaller than the two-type model suggests.

Corollary 7 (The welfare cost of bias is third order)**.** *Suppose f is continuous at $\bar{\kappa}$ with $f(\bar{\kappa}) > 0$, and that Theorem 1(ii) applies for small $\tilde{\rho}$. Then the pool has width $\bar{\kappa} - \kappa_b = 2\bar{\kappa}\tilde{\rho}/c_Q + O(\tilde{\rho}^2)$, and user welfare under the optimal mechanism falls short of the first best by*

$$W^{FB} - W_{std} \;=\; \frac{4}{3}\,\frac{f(\bar{\kappa})\,\tilde{\rho}^{\,3}}{c_Q} \;+\; O(\tilde{\rho}^{\,4}).$$

Proof. With $\bar{m}^* = c_Q/\kappa_b$, part (i) reads $\int_{\kappa_b}^{\bar{\kappa}} [c_Q(\kappa_b - t) + \tilde{\rho}\,\kappa_b] f(t)\,dt = 0$. Writing $\delta = \bar{\kappa} - \kappa_b$ and expanding f around $\bar{\kappa}$ gives $\tilde{\rho}\,\kappa_b\,\delta = c_Q\delta^2/2 + O(\delta^3)$, so $\delta = 2\bar{\kappa}\tilde{\rho}/c_Q + O(\tilde{\rho}^2)$. Users below κ_b are at their first best, and a pooled user loses $\frac{t}{2}(\bar{m}^* - c_Q/t)^2 = \frac{c_Q^2(t-\kappa_b)^2}{2\kappa_b^2\,t}$. Integrating over $[\kappa_b, \bar{\kappa}]$ gives $\frac{c_Q^2 f(\bar{\kappa})\,\delta^3}{6\bar{\kappa}^3}\,(1 + O(\delta))$, and substituting δ yields the result. □

Under the two-type menu the corresponding loss is $\tilde{\rho}^2\,\mathbb{E}[1/\kappa]/2$ (Corollary 6), which is second order. Dictating effort to an isolated type is cheap for the leader and costly for users; with a continuum, the leader can only pool, and pooling is limited to a band of terse users whose width shrinks with the bias.

Corollary 8 (The bunching point identifies the wedge)**.** *Under Theorem 1(ii), $\tilde{\rho} = \bar{m}^*\,\mathbb{E}[\kappa \mid \kappa \geq \kappa_b] - c_Q$.*

The corollary follows from part (i). It gives an empirical handle on the wedge that Section 12 notes cannot be read off pooling behavior alone: if c_Q and the distribution of κ are measured, for example through the onboarding task of Section 11, the location of the mass point in the distribution of prompt lengths recovers $\tilde{\rho}$. The standard also predicts a distinctive pattern in data: a mass point of prompt lengths at $\bar{m}^*$, a smooth density above it, and a missing mass of terse prompts below it, the pattern used to estimate responses to kinked incentives in public finance (Saez, 2010).

Remark 13 (Large bias)**.** *When the tail condition fails, the optimum adds a third tier at the terse end, where users receive partial clarification. On such a tier the first-order condition of the Lagrangian gives $m(\kappa) = \Lambda/\big(\kappa + \frac{1-r}{r}\,\frac{1-F(\kappa)}{f(\kappa)}\big)$, an inverse-hazard adjustment that equals the unassisted prompt $\Lambda/\bar{\kappa}$ at the top. For still larger bias, in our computations the most diligent users are also held to full specification, $m = k$, as type L is in Proposition 7. Figure 6 shows the first of these departures. The asking distortion and the saturation of the two-type menu thus correspond to the large-bias end of the continuum solution.*

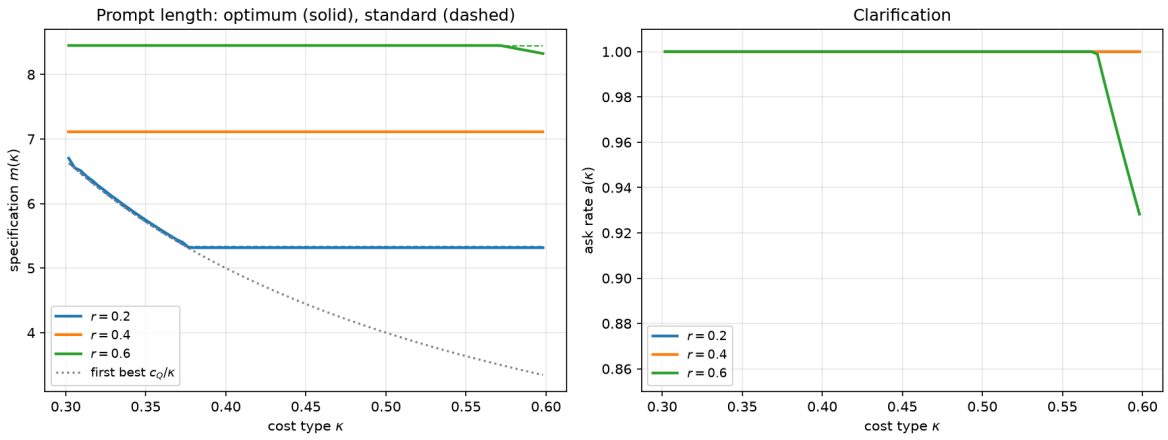


Figure 6: Optimal screening with a continuum of users, $\kappa \sim \text{Uniform}[0.3, 0.6]$, computed as the optimal mechanism for 80 equally likely types (solid) and compared with the minimum-specification standard of Theorem 1 (dashed). For $r = 0.2$ and $r = 0.4$ the tail condition holds: diligent types keep their first-best prompt c_Q/κ, the others pool at $\bar{m}^*$, and every type receives full clarification. For $r = 0.6$ it fails ($\bar{m}^* = 8.44 > \Lambda/\bar{\kappa} = 8.33$), and the tersest users receive partial clarification with slightly shorter prompts.

In the benchmark with $\kappa \sim \text{Uniform}[0.3, 0.6]$, $k = 10$, $\Lambda = 5$, $c_Q = 2$, and $\mu_A = 1$, the standard pools 46% of users at $\bar{m}^* = 4.33$ when $\lambda_A = 2.3$, and every user at $\bar{m}^* = 7.11$ when $\lambda_A = 3.2$. The user-welfare loss relative to the first best ($W^{FB} = 84.62$) is 0.05 and 1.78. A pooling assistant still asks universally at these frictions and leaves users at their first best. At $\lambda_A = 3.5$ pooling withdraws most clarification ($a^{SE} = 0.29$, $W = 78.17$), while the standard keeps everyone fully clarified at $W = 81.94$. The welfare ranking of Remark 12 therefore survives with a continuum of types.

9 Competition and Disclosure

The analysis so far has a single assistant. Suppose instead that $n \geq 2$ identical providers each post a pooling ask rate a_j and a price p_j per task, that answering costs the provider $c_A \geq 0$ per question (Remark 3), and that a unit mass of users with cost type κ choose a provider. Let $U^*(a) \equiv \max_m U(m, a; \kappa)$, $R(a) \equiv k - m^*(\kappa; a)$, and $W^c \equiv \max_{a \in [0,1]} [U^*(a) - c_A\, a\, R(a)]$, the user surplus net of the real cost of questions. Under *disclosure* users observe (a_j, p_j). Under *opacity* they observe only p_j and hold a common belief $\hat{a}$ about ask rates, either the equilibrium rate (sophisticated) or some $a^\dagger > 0$ (naive).

Proposition 10 (Competition disciplines clarification only under disclosure)**.** *(i) Under disclosure, in every pure-strategy equilibrium users obtain W^c and providers earn zero profit. The equilibrium ask rate maximizes user surplus net of the real cost of questions; the pure-bias component β_A of $\lambda_A - c_Q$ plays no role, and if $c_A = 0$ users attain the first best.*

(ii) Under opacity with $c_A > 0$, every provider that serves users sets $a_j = 0$ and $p_j = 0$. Sophisticated users choose $m^*(\kappa; 0)$; naive users choose $m^*(\kappa; a^\dagger) < m^*(\kappa; 0)$ and lose $s^2(a^\dagger)^2/(2\kappa)$ more on the interior domain.

Proof. (i) Let u be users' equilibrium utility and $\pi_j \geq 0$ provider j's profit. Total surplus at any offer is at most W^c, so $u + \sum_j \pi_j \leq W^c$. Let a^c attain W^c. A provider that deviates to $(a^c, c_A a^c R(a^c) + W^c - u - \delta)$ attracts every user and earns $W^c - u - \delta$, so $\pi_j \geq W^c - u$ for every j. With $n \geq 2$, $2(W^c - u) \leq \sum_j \pi_j \leq W^c - u$, hence $u \geq W^c$, and then $u = W^c$ and $\sum_j \pi_j = 0$. All providers offering $(a^c, c_A a^c R(a^c))$ is an equilibrium, because any offer that attracts users gives them at least W^c and so earns at most zero. (ii) Demand depends on p_j alone, and a provider's cost $c_A a_j R$ is strictly increasing in a_j whenever its users leave attributes open, so $a_j = 0$. Price competition at zero marginal cost then gives $p_j = 0$. The naive loss is Proposition 2 with $a_A = 0$. □

Clarification is a shrouded attribute in the sense of Gabaix and Laibson (2006): when users cannot see it, competition does not reward it, and the market sets it at its cost-minimizing level whatever the training objective of any single assistant. Disclosure and competition are therefore complements. Disclosure alone lets a monopolist commit to a^{SE} (Section 5); competition alone drives clarification to zero; together they deliver the ask rate a fully informed user would choose. The proposition assumes a representative user. With heterogeneous users and pooling offers, a deviating provider can attract a selected subset of users, and equilibrium existence requires the usual care (Rothschild and Stiglitz, 1976).

10 Testable Predictions

P1 (Prop. 1, Cor. 1, Section 2.3) Holding the assistant's policy fixed, higher guessing accuracy is associated with shorter specifications alongside weakly higher user welfare, and never with an overall welfare decline despite the reduction in prompt length. Under linear risk the shortening holds for every type, because $\partial^2 U / \partial m \, \partial g = -L(1-a) \leq 0$ everywhere; under the exact conjunctive payoff it is guaranteed only where Assumption 1 holds at the optimum (Remark 1).

P1′ (Model I remark) Holding accuracy fixed, the effect of a higher *ask rate* on specification length flips sign depending on whether $c_Q \gtrless \Lambda = L(1-g)$: where clarifying questions are inexpensive relative to the risk they resolve, users specify less as the ask rate rises (substitution); where questions are relatively costly to answer, users specify *more* (pre-emption). This prediction operates within the model and is distinct from P1's accuracy channel.

P2 (Prop. 2) Facing the same under-asking assistant, users who overestimate its ask rate (for example, users migrating from a high-asking assistant, or users in an opaque-policy condition) write shorter specifications and attain lower realized task success than users with accurate beliefs. The specification gap scales with $s\,(a^\dagger - a_A)/\kappa$, and, holding policies fixed, it shrinks as guessing accuracy g rises.

P3 Making ask-rate statistics visible to users shifts specification length toward the sophisticated best response analyzed in Section 3.3.

P4 (Prop. 7, Lemma 4) Under a screening policy, the clarification rate per unresolved attribute, measured as questions asked divided by the number of requirements the prompt leaves open, is weakly *lower* for terse prompts than for detailed prompts that leave attributes open: in the boundary-relaxation regime $a_H^{SB} \leq a_L^{SB} = 1$, strictly when the asking distortion is active. A policy that asks whenever an attribute is uncertain predicts equal rates. In the saturation regimes the detailed type leaves nothing open, so its rate is not observable, and only the level a_H^{SB} can be tested (P7). The raw number of questions, $a(k-m)$, rises with terseness under any policy and does not discriminate between models. Off the menu levels, any schedule that

implements the menu asks less per open attribute on prompts slightly longer than the terse level in the saturation regimes where IC_L binds (Lemma 4), so with noisy or heterogeneous prompts the screening rate is non-monotone in prompt length, whereas a pooling policy gives a flat rate.

P5 (Lemma 1) Users who do not detect all of the assistant's errors write progressively shorter specifications across an interactive session, and users with lower error-detection rates settle at shorter prompts.

P6 (Cor. 5, Cor. 6, Remark 12) Under an assistant with asking friction bias ($\lambda_A > c_Q$), a menu-screening policy achieves weakly higher leader payoff than pooling ($\Pi_{II} \geq \Pi_I$) by tailoring specification requirements to user capabilities. For user welfare ($W \equiv \mathbb{E}[U]$), however, the ranking depends on the degree of friction. While pooling still asks universally ($\lambda_A < 3.0$ in the calibration of Remark 12), users attain their first best under pooling, and screening, which dictates inefficiently long prompts, is worse for them; the gap is second order in the bias ($\tilde{\rho}^2\mathbb{E}[1/\kappa]/2$ when the leader's preferred menu is incentive compatible). Once friction pushes the pooling rate below one ($3.0 < \lambda_A < 3.5$, $a^{SE} = 2/3$) and then to zero ($3.5 \leq \lambda_A < 5.0$), screening preserves partial clarification for high-cost users and delivers higher user welfare. Under prohibitive friction ($c_Q^{eff} \geq \Lambda$) both regimes shut down asking and tie. Under the exact conjunctive payoff on long tasks, where terse users guess at their first best, an unbiased pooling policy also incurs an instrument-restriction loss by forcing a uniform rate.

P7 (Prop. 7) Even when an assistant can condition on declared specification effort, terse users receive fewer clarifying questions than the assistant's unconstrained per-type preference would prescribe ($a_H^{SB} < a_H^B$). This downward distortion arises specifically from the incentive-compatibility constraint IC_L. In the low-cost saturation regime ($\kappa_L \leq \bar{\kappa}_L$), downward distortion intensifies as cost heterogeneity narrows (κ_L approaches $\bar{\kappa}_L$), requiring deeper distortion to prevent mimicry; in the boundary relaxation regime ($\bar{\kappa}_L < \kappa_L < c_Q^{eff}/k$), the assistant grants clarification to low-cost users ($a_L^{SB} = 1$), relaxing IC_L and strictly dampening downward distortion on high-cost users relative to the saturated formula (preventing the steep collapse that would occur if a_L were held at 0).

P8 (Prop. 5) Under an assistant with under-asking friction bias ($\lambda_A > c_Q$), higher model guessing capability g can perversely lengthen user prompts ($d\mathbb{E}[m^*]/dg > 0$) and lower user welfare ($dW/dg < 0$). This equilibrium capability reversal occurs if and only if the assistant's ask-rate reduction is sufficiently steep to outpace direct guessing gains ($-\frac{da^{SE}}{dg} > \frac{L(1-a^{SE})}{\Lambda(g)-c_Q}$), forcing users to over-specify upstream to defensively insure against guessing errors.

P9 (Theorem 1, Corollary 8) An assistant that conditions clarification on prompt completeness, facing users who differ continuously in specification cost, produces a mass point in the distribution of prompt completeness at the standard $\bar{m}^*$, missing mass just below it, and a jump in the per-attribute ask rate to one at $\bar{m}^*$. Users who write long prompts anyway are unaffected. The location of the mass point, together with measured specification costs, identifies the wedge $\tilde{\rho}$.

P10 (Proposition 10) Where assistants compete, publishing ask rates raises clarification toward the level users would choose; without disclosure, competition lowers clarification rather than raising it.

11 Empirical Strategy and Experimental Design

The behavioral predictions P1–P10 (including P1′) translate our contract-theoretic mechanisms into observable margins for experimental testing. In naturalistic software engineering logs, however, assistant policy bias λ_A and user specification costs κ are typically confounded with task difficulty and user experience. To causally test our predictions, we propose a between-subjects randomized controlled trial on software developers performing standardized implementation tasks:

Experimental design and treatments. Participants are randomly assigned to interact with code assistants operating under distinct clarification regimes:

1. *Treatment A (Low-asking policy, $a = 0.1$):* The assistant aggressively guesses ambiguous attributes with minimal clarification.
2. *Treatment B (High-asking policy, $a = 0.8$):* The assistant frequently asks targeted clarifying questions before generating code.
3. *Treatment C (Menu-conditioned screening policy):* The assistant sets its clarification rate from a real-time measure of how complete the initial prompt is, implementing the minimum-specification standard of Theorem 1 through a continuous schedule (Section 8).
4. *Control (Unbiased benchmark):* The assistant asks about every residual ambiguity it detects ($a = 1$), the first-best policy of Proposition 3 under linear risk.

Crossed with the policy arms, a second randomized factor varies *policy disclosure*. In the opaque condition, participants are told nothing about the assistant's asking behavior; in the transparent condition, they see the arm's realized ask rate before the first task. This 4×2 design separates the belief channel of Proposition 2 (P2, P3) from the substitution channel (P1, P1′). Within the low-asking arm, it also identifies naive under-specification as the opaque–transparent difference in prompt length and task success.

Incentives and parameter control. A flat hourly fee alone would leave the private loss from a defect near zero, or driven only by intrinsic motivation, and would attenuate every prediction that runs through the stake Λ. Participants should therefore receive a base fee plus a bonus tied to hidden tests, one per target requirement. Paying L for each requirement the delivered code satisfies induces the additive linear-risk payoff with a known per-attribute stake; paying a single bonus V only when every test passes induces the conjunctive payoff. Randomizing the bonus scheme therefore also tests the curvature distinction of Section 12. A fixed deduction from the bonus for each clarifying question answered adds a known component to c_Q. Treatment C needs m before the assistant decides whether to ask: a classifier, validated against human coding on held-out prompts, counts how many of the task's k pre-registered requirements the initial prompt specifies. Because this measure is noisy, the continuous schedule matters in practice: under the discontinuous schedule of Lemma 3, a one-requirement misclassification would switch clarification off entirely.

Measurement, identification, and power analysis. For each programming session, instrumentation logs: (i) upstream prompt length and specified invariants/edge cases (m); (ii) clarification turn count ($a(k - m)$); (iii) downstream test suite execution and correctness; and (iv) post-session subjective developer satisfaction. By exogenously varying the clarification policy a, this protocol directly tests whether users economize on prompt specification in response to anticipated assistance ($m^*(a)$ effort substitution, Prediction P1), distinguishes the substitution channel

($\Lambda > c_Q$) from the pre-emption channel ($c_Q > \Lambda$, Prediction P1′), and evaluates whether the per-attribute clarification rate depends on user specification terseness (Prediction P4).

To identify user specification cost parameters κ without confounding from unobserved task difficulty, the protocol includes a standardized pre-trial onboarding task where participants articulate requirements without assistant interaction, establishing an individual-level baseline specification effort per unit of task complexity. Assistant asking friction λ_A is exogenously manipulated through provider system prompt instructions, conversational turn penalties, and backend latency budgets. The same baseline measure makes P9 testable: in Treatment C, the distribution of prompt completeness should show a mass point at the standard, and Corollary 8 converts its location and the baseline costs of the users who pool there into an estimate of $\tilde{\rho}$, which can be compared with the friction the experimenter imposed. A power calculation for the primary contrast (prompt length m, low-asking versus high-asking arm, two-sided two-sample t-test) shows that detecting a small-to-medium effect of $d = 0.35$ standard deviations at $\alpha = 0.05$ with 80% power requires $n = 130$ developers per policy arm, or 520 across the four arms. If all six pairwise arm contrasts are treated as confirmatory with a Bonferroni correction ($\alpha = 0.05/6$), the requirement rises to $n = 200$ per arm. Because the disclosure factor is crossed within arms, its main effect is estimated on the full sample; the opaque–transparent contrast within the low-asking arm alone has power 0.51 at $d = 0.35$ with $n = 130$ per arm (about 65 per cell), so testing P2/P3 within that arm requires either a larger arm or a pre-registered one-sided test. Because each developer completes several tasks, within-subject repetition and task fixed effects further reduce residual variance. We therefore recommend pre-registering the primary contrast, the d assumption, and the correction method.

Ethics and human subjects considerations. The proposed human evaluation involves non-invasive coding tasks. Participating developers should be recruited through professional developer platforms, provide explicit informed consent, and receive a base fee at standard market rates for professional software engineering (e.g., \$50–\$75 per hour) plus the performance bonus described above. All logged code, prompts, and telemetry must be fully anonymized prior to analysis, with no personally identifiable information recorded. The study protocol should be submitted to the relevant Institutional Review Board (IRB) for formal exemption or expedited review under institutional guidelines for research involving benign behavioral interventions. To facilitate reproducibility and protocol validation prior to live human deployment, we provide a companion open-source simulation harness (Appendix D) supporting both deterministic offline dry-run evaluation with mock LLM client responses and live API interactions across pooling and screening treatments.

12 Discussion and Limitations

The linear-risk formulation is the main tractability baseline analyzed in Sections 4–7. Rather than viewing linear risk solely as a small-risk Taylor approximation, it is properly understood as an independent, additive-defect primitive model:

$$U(m, a; \kappa) \;=\; V \;-\; L(1-q)(k-m) \;-\; \frac{\kappa}{2}m^2 \;-\; a\, c_Q\,(k-m),$$

where L represents the localized defect cost per unresolved attribute and cumulative defect loss satisfies $kL \leq V$. In this additive environment, each unaddressed attribute produces an independent defect without destroying the entire project. By contrast, the exact conjunctive specification $Vq(a, g)^{k-m}$ represents a systemic bottleneck environment where all attributes are strictly complementary and a single error destroys the entire deliverable. To calibrate the exact model against the linear baseline, note that for benchmark task size $k = 10$, total project valuation $V = 100$

corresponds precisely to cumulative defect loss $kL = 100$ (with per-attribute stake $L = 10$). To investigate the role of payoff curvature, we evaluate both the unbiased pooling optimum and the biased screening menu directly under the exact conjunctive payoff $q(a, g)^{k-m}$ via direct numerical grid search and constrained optimization. The exact specification is a distinct environment, and the comparisons below show where its curvature matters and where it does not.

Under this exact nonlinear specification across task complexity $k \in [3, 20]$ and guessing accuracy $g \in [0.50, 0.95]$ (30 grid points, $\kappa \sim \text{Uniform}[0.2, 0.6]$, $c_Q = 2$):

1. The corner-solution property of Corollary 4 persists: the unbiased pooling optimum is $a^{SE} = 1$ in all 25 configurations with $k \geq 5$. In the 5 configurations with $k = 3$ every type specifies all three attributes under both $a = 0$ and $a = 1$, so the payoff is flat in a and asking is payoff-irrelevant.

2. Unbiased pooling choices agree with the linear Taylor-expansion baseline ($L = V = 100$) in every configuration in which either model has a strict preference; the 5 apparent disagreements at $k = 3$ are the ties just described, broken in opposite directions by the two solvers. Against the additive-defect baseline ($L = 10$) there are 10 strict divergences, all at $g \geq 0.90$, where $\Lambda = L(1 - g) \leq 1 < c_Q$. The linear model then rationally guesses, outside the maintained assumption $\Lambda > c_Q$, while the exact model still asks, because its per-question stake $V(1-g) \geq 5$ exceeds c_Q. Neither comparison produces a zero-bias gap between pooling and screening.

3. A zero-bias gap does arise under the exact payoff, but only on long tasks. In 1,500 random two-type populations ($k \in \{3, \ldots, 30\}$, $g \in [0.5, 0.98]$, $c_Q \in [0.5, 6]$, $V = 100$), the gap $\mathbb{E}[\max_{m,a} U] - \max_a \mathbb{E}[\max_m U]$ is positive in 61 draws (4.1%), in none of the 517 draws with $k \leq 12$, and is at most 5.2. In every one of the 61 draws the terse type guesses at its first best and the diligent type asks, and the terse type's question bill under full asking, $c_Q(k - m)$, is never below $0.49V$ (median about V). This matches Remark 5: the per-question value of asking is highest when few attributes remain, so guessing is first best only for a user facing many open attributes. By the envelope theorem, the gain from asking, $G(\kappa) = \max_m U(m, 1; \kappa) - \max_m U(m, 0; \kappa)$, satisfies $G'(\kappa) = [m_0(\kappa)^2 - m_1(\kappa)^2]/2$, so it falls with κ wherever users specify more under asking than under guessing; this holds for the terse type in all 61 draws.

4. Downward screening distortion ($a_H^{SB} < a_H^B$) holds in 20 of 30 cases (67% overall: 20% at $k \in \{3, 5\}$, 80% at $k \in \{8, 20\}$, and 100% at $k \in \{10, 15\}$). The low incidence of distortion at small k stems from boundary saturation: when tasks are simple ($k \leq 5$), users can fully specify all attributes ($m = k$), leaving no residual ambiguity for clarification to resolve ($k - m = 0$). Consequently, both IC_L and IC_H are active in 15 of 30 cases, concentrated entirely at small task sizes ($k \leq 8$).

Furthermore, numerical solutions of the full four-variable constrained menu problem under the exact payoff across $\lambda_A \in [c_Q, 10c_Q]$ and $\mu_A \in [0.1, 1.0]$ (Figure 7) illustrate the boundary transition to clarification shutdown. In the linear model with $\mu_A = 1.0$, asking friction reaches or exceeds task stakes ($c_Q^{eff} \geq \Lambda$) at $\lambda_A \geq 5.0$. Beyond this threshold, the leader unconstrainedly prefers zero asking for all types ($a_L^B = a_H^B = 0$). Under the exact nonlinear payoff, this technological clarification shutdown threshold shifts outward to $\lambda_A \approx 8.0$. In this shutdown regime, both types choose full specification $m = k$ and receive $(k, 0)$, so IC_H and IC_L both hold with equality ($0 = 0$). This co-activity is not an adverse-selection screening distortion, but a technological shutdown artifact where clarification is universally priced out by severe asking friction.

Another limitation concerns the off-menu schedule. Lemma 3 implements Proposition 7 with a policy that asks only at the menu levels, so the menu does not rely on users being restricted

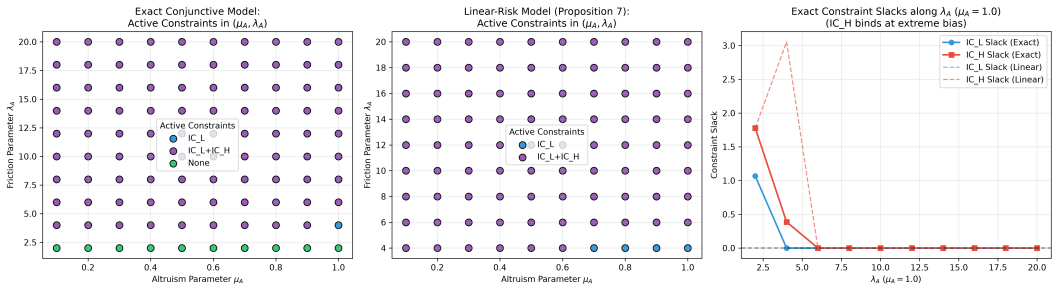


Figure 7: Screening menu allocations and active constraint boundaries under the exact conjunctive payoff across assistant asking friction λ_A and weight μ_A, illustrating the transition to clarification shutdown at $\lambda_A \approx 8.0$.

to the listed bundles. Lemma 4 shows that this discontinuity is not needed: the most generous implementing schedule is continuous in prompt length, so a small error in measuring m changes the ask rate only slightly. Non-monotonicity cannot be removed. A natural minimum-effort rule, asking whenever the prompt is at least m_H^{SB} long, fails in the saturation regimes where IC_L binds (regimes 2 and 3 of Proposition 7) and in Corollary 6, because the diligent type would shorten its prompt and keep the clarification. A schedule that asks less on somewhat more complete prompts may seem arbitrary to users. Whether it survives prompt gaming, for example padding a prompt with detail that does not resolve any target attribute, depends on how well m is measured, and is a question for applied mechanism design.

The menu also requires commitment, and it is not renegotiation-proof. Once a user has revealed type H by choosing m_H^{SB}, the leader's own payoff is strictly increasing in a_H (Step 1 of the proof in Appendix A), so ex post it would rather ask more than a_H^{SB}. Likewise, the off-menu responses, whether $a = 0$ or the continuous schedule of Lemma 4, ask less than the leader would want to ex post, since $c_Q^{eff} < \Lambda$ makes asking profitable for the leader at any $m < k$. Commitment is more plausible for an AI assistant than for a human principal, because the policy is fixed at training or deployment time and applied mechanically to every prompt. A provider that re-tunes the policy after observing the prompt distribution, however, faces the familiar ratchet problem: the distortion that sustains separation is exactly what it would want to undo.

Regarding the regularity condition (Assumption 1), decreasing differences on the entire domain $m \in [0, k]$ (the hypothesis of Topkis's theorem) reduces to $k \ln q \geq -1$, which holds in 24.8% of a broad numerical parameter grid. For local comparative statics at a smooth interior optimum, the sign condition need only hold at m^* (Remark 1). The local condition $(k - m^*) \ln q \geq -1$ holds in 89.9% of the grid. A finite-difference audit finds that $m^*(g)$ is weakly decreasing in all 2,162 grid intervals where the local condition holds. In the 390 intervals where it fails, monotonicity breaks in 377 (96.7%): higher guessing accuracy raises specification effort, and the 13 survivors are boundary-saturated at $m^* = k$. The local condition is therefore close to necessary as well as sufficient. Under the exact conjunctive payoff, Corollary 1 should be read as a statement about the empirically relevant region where users leave few attributes unspecified, not as a global law. Under linear risk no such caveat is needed (Section 2.3).

The analytically tractable model is, by design, degenerate at zero bias. Under linear risk an unbiased assistant asks about every residual attribute, pooling and screening coincide, and incentive compatibility is free (Propositions 3 and 6, Corollary 4). Every non-trivial result in Sections 5–7 is therefore driven by the wedge $\tilde{\rho}$: the paper is a theory of how a misaligned asking cost propagates through users' specification choices, not of why asking and guessing trade off at the first best. The

exact payoff produces a first-best trade-off only on long tasks (item 3 of the exact-model comparison above), and ordered attribute stakes produce one with the opposite sorting (Proposition 8); in both cases the zero-bias gap is small.

A stylized modeling assumption is attribute symmetry (k binary attributes with identical loss L and guessing probability g). In practical software engineering, attributes exhibit substantial loss heterogeneity: omitting an architectural invariant or security convention causes severe failure ($L_i \gg 0$), while omitting a stylistic naming convention carries negligible cost ($L_j \approx 0$). Section 7.1 relaxes this under linear risk: users specify the highest-stakes attributes first, and with an attribute-blind ask rate the first best sorts users by a threshold in κ (Proposition 8). Two extensions remain open. The first is screening under bias when stakes differ, where the menu could condition on which attributes a prompt leaves open as well as on how many. The second is heterogeneity in guessing accuracy g_i across attributes, which changes the order in which users specify. We conjecture that the core mechanism, the substitution between upstream specification and clarification under misaligned asking friction, carries over to both.

A related limitation concerns the empirical identification of λ_A. Every bias-driven prediction (Corollary 2, Proposition 7, and predictions P2, P6, P7, and P8) is stated in terms of under-asking bias ($\lambda_A > c_Q$), but backing out λ_A from observed policy alone is circular. Lemma 2 formalizes this, within the multiplicative specification (Remark 7), as an exact non-identification result: a^{SE} depends on (μ_A, λ_A) only through their ratio $\tilde{\rho} = (\lambda_A - c_Q)/\mu_A$, so a discounting-driven leader and a friction-driven leader with the identical excess friction ratio cannot be distinguished from pooling behavior alone. An independent empirical handle, such as an explicit penalty on turn count in the training objective or reward specification, would be needed before the bias-versus-heterogeneity predictions in Section 10 can be tested without circularity. Corollary 8 supplies one such handle when the assistant screens a continuum of users. Theorem 1 characterizes that case under a tail condition; a full characterization when the condition fails, including the order in which the tiers of Remark 13 appear, remains open. We also assume truthful revelation of specified attributes (abstracting from cheap-talk ambiguity within what is communicated). Section 9 treats competition with a representative user; competition between screening menus with heterogeneous users is left for future work.

A broader methodological conclusion concerns the limitations of applying classical contract-theoretic intuition directly to human–AI settings without accounting for non-monetary, partially aligned incentives. In classical mechanism design (such as Baron–Myerson, Laffont–Tirole, or costly verification models (Baron and Myerson, 1982; Laffont and Tirole, 1986; Ben-Porath et al., 2014)), the principal pays monetary transfers to extract surplus, and agent informational rent directly reduces the principal's payoff. In human–AI interaction, the assistant's objective directly incorporates user welfare. As a consequence, incentive compatibility is costless in the absence of bias (Proposition 6). Under bias, the rent pattern is reversed: the efficient specifier is held to its outside option while the distorted high-cost type retains a rent (Remark 10). And unbiased pooling induces corner solutions rather than interior compromises (Corollary 4). With a continuum of users the departure is starker still: the principal values the transfer it makes, so under moderate bias it never withholds clarification from users who meet its standard, and the entire cost of misalignment takes the form of pooled effort (Theorem 1). Economic analyses of automated assistance must therefore explicitly incorporate the non-monetary, partially aligned nature of the interaction rather than relying on direct analogies to cash-transfer mechanisms.

References

Philippe Aghion and Jean Tirole. Formal and real authority in organizations. *Journal of Political Economy*, 105(1):1–29, 1997.

Mohammad Aliannejadi, Hamed Zamani, Fabio Crestani, and W. Bruce Croft. Asking clarifying questions in open-domain information-seeking conversations. In *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval* (SIGIR), pages 475–484, 2019.

Ricardo Alonso and Niko Matouschek. Optimal delegation. *Review of Economic Studies*, 75(1):259–293, 2008.

Attila Ambrus and Georgy Egorov. Delegation and nonmonetary incentives. *Journal of Economic Theory*, 171:101–135, 2017.

Mark Armstrong and John Vickers. A model of delegated project choice. *Econometrica*, 78(1):213–244, 2010.

David P. Baron and Roger B. Myerson. Regulating a monopolist with unknown costs. *Econometrica*, 50(4):911–930, 1982.

Elchanan Ben-Porath, Eddie Dekel, and Barton L. Lipman. Optimal allocation with costly verification. *American Economic Review*, 104(12):3779–3813, 2014.

Vincent P. Crawford and Joel Sobel. Strategic information transmission. *Econometrica*, 50(6):1431–1451, 1982.

Wouter Dessein. Authority and communication in organizations. *Review of Economic Studies*, 69(4):811–838, 2002.

Nicholas Edwards and Sebastian Schuster. Ask or assume? Uncertainty-aware clarification-seeking in coding agents. *arXiv preprint arXiv:2603.26233*, 2026.

Xavier Gabaix and David Laibson. Shrouded attributes, consumer myopia, and information suppression in competitive markets. *Quarterly Journal of Economics*, 121(2):505–540, 2006.

Sanford J. Grossman. The informational role of warranties and private disclosure about product quality. *Journal of Law and Economics*, 24(3):461–483, 1981.

Dylan Hadfield-Menell, Anca Dragan, Pieter Abbeel, and Stuart Russell. Cooperative inverse reinforcement learning. In *Advances in Neural Information Processing Systems* (NIPS), volume 29, pages 3909–3917, 2016.

Dylan Hadfield-Menell and Gillian K. Hadfield. Incomplete contracting and AI alignment. In *Proceedings of the 2019 AAAI/ACM Conference on AI, Ethics, and Society* (AIES), pages 417–422, 2019.

Bengt Holmström. On the theory of delegation. In M. Boyer and R. E. Kihlstrom, editors, *Bayesian Models in Economic Theory*, pages 115–141. North-Holland, Amsterdam, 1984.

Emir Kamenica and Matthew Gentzkow. Bayesian persuasion. *American Economic Review*, 101(6):2590–2615, 2011.

Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. CLAM: Selective clarification for ambiguous questions with generative language models. *arXiv preprint arXiv:2212.07769*, 2022. Workshop version in *ICML 2023 Workshop on Challenges in Deployable Generative AI*, 2023.

Philippe Laban, Hiroaki Hayashi, Yingbo Zhou, and Jennifer Neville. LLMs get lost in multi-turn conversation. *arXiv preprint arXiv:2505.06120*, 2025.

Jean-Jacques Laffont and Jean Tirole. Using cost observation to regulate firms. *Journal of Political Economy*, 94(3):614–641, 1986.

Paul Milgrom and Ilya Segal. Envelope theorems for arbitrary choice sets. *Econometrica*, 70(2):583–601, 2002.

Paul Milgrom and Christina Shannon. Monotone comparative statics. *Econometrica*, 62(1):157–180, 1994.

Sewon Min, Julian Michael, Hannaneh Hajishirzi, and Luke Zettlemoyer. AmbigQA: Answering ambiguous open-domain questions. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing* (EMNLP), pages 5783–5797, 2020.

James A. Mirrlees. An exploration in the theory of optimum income taxation. *Review of Economic Studies*, 38(2):175–208, 1971.

Michael Mussa and Sherwin Rosen. Monopoly and product quality. *Journal of Economic Theory*, 18(2):301–317, 1978.

Roger B. Myerson. Incentive compatibility and the bargaining problem. *Econometrica*, 47(1):61–73, 1979.

Roger B. Myerson. Optimal auction design. *Mathematics of Operations Research*, 6(1):58–73, 1981.

Sudha Rao and Hal Daumé III. Learning to ask good questions: Ranking clarification questions using neural expected value of perfect information. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics* (ACL), pages 2737–2746, 2018.

Michael Rothschild and Joseph Stiglitz. Equilibrium in competitive insurance markets: An essay on the economics of imperfect information. *Quarterly Journal of Economics*, 90(4):629–649, 1976.

Emmanuel Saez. Do taxpayers bunch at kink points? *American Economic Journal: Economic Policy*, 2(3):180–212, 2010.

Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvenaud, Amanda Askell, Samuel R. Bowman, Newton Cheng, Esin Durmus, Zac Hatfield-Dodds, Scott R. Johnston, Shauna Kravec, Timothy Maxwell, Nicholas Joseph, Liane Lovitt, Sam McCandlish, Kamal Ndousse, Nicholas Schiefer, Andy Zou, Julian Michael, and Ethan Perez. Towards understanding sycophancy in language models. *arXiv preprint arXiv:2310.13548*, 2023.

Christopher A. Sims. Implications of rational inattention. *Journal of Monetary Economics*, 50(3):665–690, 2003.

Prasann Singhal, Tanya Goyal, Jiacheng Xu, and Greg Durrett. A long way to go: Investigating length correlations in RLHF. *arXiv preprint arXiv:2310.03716*, 2023.

Michael Spence. Job market signaling. *Quarterly Journal of Economics*, 87(3):355–374, 1973.

Donald M. Topkis. Minimizing a submodular function on a lattice. *Operations Research*, 26(2):305–321, 1978.

Donald M. Topkis. *Supermodularity and Complementarity*. Princeton University Press, Princeton, NJ, 1998.

Robert M. Townsend. Optimal contracts and competitive markets with costly state verification. *Journal of Economic Theory*, 21(2):265–293, 1979.

Sanidhya Vijayvargiya, Xuhui Zhou, Akhila Yerukola, Maarten Sap, and Graham Neubig. Ambig-SWE: Interactive agents to overcome underspecificity in software engineering. *arXiv preprint arXiv:2502.13069*, 2025.

Heinrich von Stackelberg. *Marktform und Gleichgewicht*. Julius Springer, Berlin/Wien, 1934.

Chenyang Yang, Yike Shi, Qianou Ma, Michael Xieyang Liu, Christian Kästner, and Tongshuang Wu. What prompts don't say: Understanding and managing underspecification in LLM prompts. *arXiv preprint arXiv:2505.13360*, 2025.

Hamed Zamani, Susan Dumais, Nick Craswell, Paul Bennett, and Gord Lueck. Generating clarifying questions for information retrieval. In *Proceedings of the Web Conference 2020* (WWW), pages 418–428, 2020.

A Proof of Proposition 7

Proof of Proposition 7. We establish the result in five analytical steps:

Step 1: Unconstrained biased optima. The leader objective per type is $\Pi_\kappa(m,a) = \mu_A[V - \Lambda(1-a)(k-m) - \frac{\kappa}{2}m^2 - ac_Q^{eff}(k-m)]$, where $c_Q^{eff} \equiv c_Q + (\lambda_A - c_Q)/\mu_A$. Under the maintained condition $0 < \lambda_A - c_Q < \mu_A(\Lambda - c_Q)$, effective asking friction satisfies $c_Q < c_Q^{eff} < \Lambda$. For type H with $\kappa_H > c_Q^{eff}/k$, because $c_Q^{eff} < \Lambda$, Π_{κ_H} is strictly increasing in a for any $m < k$ (slope $\mu_A(\Lambda - c_Q^{eff})(k-m) > 0$). The leader's unconstrained optimum for type H therefore sets $a_H^B = 1$ and interior specification $m_H^B = c_Q^{eff}/\kappa_H < k$. For type L with $\kappa_L < c_Q^{eff}/k$, the partial derivative with respect to effort satisfies $\partial\Pi_{\kappa_L}/\partial m = \mu_A[\Lambda(1-a) + ac_Q^{eff} - \kappa_L m] \geq \mu_A[c_Q^{eff} - \kappa_L k] > 0$ for all $m \in [0,k]$ and all $a \in [0,1]$. Hence Π_{κ_L} is strictly increasing in m, and attains its unconstrained maximum on $[0,k]\times[0,1]$ at $m_L^B = k$. At $m_L^B = k$, zero attributes remain unspecified ($k - m_L^B = 0$), so asking is payoff-irrelevant to both the user and the leader ($\Pi_{\kappa_L}(k,a) = \mu_A[V - \frac{\kappa_L}{2}k^2]$ for all a). Setting $a_L^B = 0$ is the unconstrained benchmark choice because it gives zero asking service, minimizing the temptation of type H to mimic type L.

Step 2: Upward temptation to mimic (IC_L violation). Evaluating type κ_L's payoff from mimicking the high type's unconstrained bundle $(m_H^B, 1)$ versus accepting its own bundle $(k, 0)$:

$$U(m_H^B, 1; \kappa_L) - U(k, 0; \kappa_L) \;=\; (k - m_H^B)\left[\frac{\kappa_L(k + m_H^B)}{2} - c_Q\right].$$

Because $m_H^B < k$, this difference is strictly positive if and only if $\kappa_L(k + m_H^B) > 2c_Q$, which is equivalent to $k > k_{\text{slack}} \equiv \frac{2c_Q}{\kappa_L} - m_H^B = \frac{2c_Q}{\kappa_L} - \frac{c_Q^{eff}}{\kappa_H}$. When $k \leq k_{\text{slack}}$, IC_L is strictly satisfied at the unconstrained bundles, establishing claim 1.

Step 3: Relaxed problem and slackness of dropped constraints (IC_H, IR_H). We formulate the relaxed screening problem by maximizing the expected leader objective subject to IC_L alone, dropping IC_H, IR_L, and IR_H. We then establish that all dropped constraints are strictly satisfied at the relaxed optimum:

1. *Slackness of IC_H via single-crossing:* By the Spence–Mirrlees sorting condition $\frac{\partial^2 U}{\partial m\,\partial \kappa} = -m < 0$, the rent identity for any menu $((m_L, a_L), (m_H, a_H))$ is:

$$U(m_H, a_H; \kappa_H) - U(m_L, a_L; \kappa_H) \;=\; [U(m_H, a_H; \kappa_L) - U(m_L, a_L; \kappa_L)] \;+\; \frac{\kappa_H - \kappa_L}{2}\big(m_L^2 - m_H^2\big).$$

When $(m_L, a_L) = (k, 0)$ and IC_L binds with equality ($U(m_H, a_H; \kappa_L) = U(k, 0; \kappa_L)$), this simplifies directly to:

$$U(m_H, a_H; \kappa_H) - U(k, 0; \kappa_H) \;=\; \frac{\kappa_H - \kappa_L}{2}\big(k^2 - m_H^2\big) \;>\; 0,$$

which is strictly positive for any $m_H < k$ since $\kappa_H > \kappa_L$. Hence IC_H holds with strict inequality (strict slack).

2. *Participation:* With $\underline{U} = 0$ and V large, both IR constraints are slack. Against the stronger endogenous outside option $\underline{U}(\kappa) = \max_m U(m, 0; \kappa)$, Lemma 3 shows that both types weakly prefer their bundles; IR_L holds with equality whenever $(m_L, a_L) = (k, 0)$, and IR_H holds strictly whenever $m_H^{SB} < m^o$.

Step 4: Global optimization along the IC_L boundary and regimes. Because Π_{κ_H} is strictly increasing in a_H on $c_Q^{eff} < \Lambda$, the leader pushes a_H as high as incentive compatibility permits, so IC_L must bind at the relaxed optimum. Along the binding boundary $U(m_H, a_H; \kappa_L) = U_L^0 \equiv V - \frac{\kappa_L}{2}k^2$, utility is strictly linear in a_H, uniquely determining:

$$a_H(m_H) \;=\; 1 \;-\; \frac{U(m_H, 1; \kappa_L) - U_L^0}{s(k - m_H)} \;=\; 1 \;-\; \frac{\frac{\kappa_L(k+m_H)}{2} - c_Q}{s}.$$

Substituting this boundary into the leader's objective $\Pi_{\kappa_H}(m_H, a_H(m_H))$ yields a strictly concave quadratic in m_H:

$$\frac{d^2\Pi_{\kappa_H}}{dm_H^2} \;=\; -\,\frac{\mu_A\big[s(\kappa_H - \kappa_L) + \tilde{\rho}\kappa_L\big]}{s} \;<\; 0,$$

which attains its unique unconstrained stationary point at:

$$m_H^* \;=\; \frac{b\Lambda}{\mu_A s(\kappa_H - \kappa_L) + b\kappa_L} \;=\; \frac{\Lambda\tilde{\rho}}{s(\kappa_H - \kappa_L) + \tilde{\rho}\kappa_L}.$$

Now consider the interaction between the constraints $m_H \le k$ and $a_H \le 1$:

- If $k_{\text{slack}} < k \le k_{\text{crit}} \equiv \frac{2c_Q}{\kappa_L} - m_H^*$ (which is non-empty whenever $m_H^* < m_H^B$), then $\frac{\kappa_L(k+m_H^*)}{2} \le c_Q$. Evaluating $a_H(m_H^*)$ would yield $a_H \ge 1$. Since $a_H \le 1$, the upper bound $a_H = 1$ binds. At $a_H = 1$, IC_L reads $(k - m_H)[\frac{\kappa_L(k+m_H)}{2} - c_Q] \le 0$, i.e., $m_H \le \hat{m} \equiv \frac{2c_Q}{\kappa_L} - k$, and $\hat{m} < m_H^B$ because $k > k_{\text{slack}}$. On $m_H \le \hat{m}$, $\Pi_{\kappa_H}(m_H, 1)$ is increasing (as $\hat{m} < m_H^B$); on $m_H > \hat{m}$, the leader is on the boundary curve $a_H(m_H) < 1$, along which Π_{κ_H} is concave with peak $m_H^* \le \hat{m}$ and hence decreasing. The maximizer is therefore the kink $m_H = \hat{m}$ with $a_H = 1$. The distortion manifests entirely through prompt effort, establishing claim 2. When $m_H^* \ge m_H^B$, $k_{\text{slack}} \ge k_{\text{crit}}$ and this intermediate effort-distortion regime is empty.

- If $k > \max\{k_{\text{slack}}, k_{\text{crit}}\}$ and $k \geq m_H^*$, then $\frac{\kappa_L(k+m_H^*)}{2} > c_Q$, yielding interior asking distortion $a_H^{SB} = a_H(m_H^*) < 1$ with effort $m_H^{SB} = m_H^*$, establishing claim 3.
- If $k < m_H^*$ (which occurs when cost heterogeneity $\kappa_H - \kappa_L$ is small), the stationary point exceeds the physical attribute count ($m_H^* > k$). By strict concavity of $\Pi_{\kappa_H}(m_H)$ on $[0, k]$, the constrained maximizer is $m_H^{SB} = k$. At $m_H = k$, residual attributes vanish ($k - m_H = 0$), so asking is payoff-irrelevant and both types specify fully, establishing claim 4.

We now determine when $(m_L^{SB}, a_L^{SB}) = (k, 0)$ is optimal. Consider departing from $m_L = k$ by offering type L clarification service ($a_L = 1$) with $m_L < k$. Since $\kappa_L > c_Q/k$, reducing m_L below k increases type L's utility at marginal rate $\frac{\partial U}{\partial m_L}\Big|_{m_L=k} = c_Q - \kappa_L k < 0$, yielding a utility increase of $\kappa_L k - c_Q > 0$. With population shares f_L, f_H, increasing type L's utility by $+1$ relaxes IC_L, allowing a_H to increase by $\frac{1}{s(k-m_H)}$, which increases the leader's payoff on type H at rate $f_H \frac{\mu_A s - b}{s}(\kappa_L k - c_Q)$. Conversely, granting type L clarification costs the leader $f_L \mu_A (c_Q^{eff} - \kappa_L k)$ per unit of reduction in m_L. Departing from $(k, 0)$ is strictly unprofitable if and only if the marginal gain on type H is weakly less than the marginal cost on type L:

$$f_H \frac{\mu_A s - b}{s}(\kappa_L k - c_Q) \;\leq\; f_L \mu_A (c_Q^{eff} - \kappa_L k).$$

Dividing by $f_L \mu_A$ and substituting $\tilde{\rho} = b/\mu_A$ and $t \equiv \frac{f_H}{f_L}\frac{s-\tilde{\rho}}{s}$, this simplifies to:

$$t(\kappa_L k - c_Q) \;\leq\; c_Q^{eff} - \kappa_L k \iff \kappa_L \;\leq\; \bar{\kappa}_L \;\equiv\; \frac{c_Q^{eff} + t c_Q}{k(1+t)}.$$

When $f_L = f_H = 0.5$, $t = \frac{s-\tilde{\rho}}{s}$ and $\bar{\kappa}_L$ simplifies to $\frac{s c_Q^{eff} + (s-\tilde{\rho}) c_Q}{k(2s-\tilde{\rho})}$. This local condition is also global in regimes 3–4. Any departure with $m_L < k$ optimally sets $a_L = 1$ (both U_L and Π_{κ_L} increase in a_L when $c_Q^{eff} < \Lambda$). Along the boundary, Π_{κ_H} is affine in the promised utility U_L with constant slope $\mu_A(s-\tilde{\rho})/s$, because $(1 - a_H)(k - m_H) = [U(m_H, 1; \kappa_L) - U_L]/s$ and m_H^* does not depend on U_L. Meanwhile $U(m_L, 1; \kappa_L)$ and $\Pi_{\kappa_L}(m_L, 1)$ are concave in m_L. The leader's objective is therefore concave in m_L, and the first-order comparison at $m_L = k$ is necessary and sufficient. Hence, when $\kappa_L \leq \bar{\kappa}_L$, $(m_L^{SB}, a_L^{SB}) = (k, 0)$ is the second-best optimum; when $\bar{\kappa}_L < \kappa_L < c_Q^{eff}/k$, the leader departs from $(k, 0)$ by setting $a_L^{SB} = 1$ and $m_L^{SB} < k$, establishing claim 5.

In the effort-distortion regime, the shadow value of relaxing IC_L is different, because the leader uses the extra slack to raise m_H along $a_H = 1$ rather than to raise a_H. Raising U_L by one unit raises m_H by $1/(\kappa_L k - c_Q)$ (the slope of $U(\cdot, 1; \kappa_L)$ at $\hat{m}$), and each unit of m_H is worth $\mu_A(c_Q^{eff} - \kappa_H \hat{m}) > 0$ to the leader. Repeating the marginal comparison gives the saturation condition $f_H(c_Q^{eff} - \kappa_H \hat{m}) \leq f_L(c_Q^{eff} - \kappa_L k)$ stated in claim 2. Here the leader's value is no longer affine in U_L, so this first-order condition is necessary. An exact grid solver confirms that it is also sufficient: in all 82 of 120 random effort-regime draws in which it holds, no departure is profitable. The same solver matches the closed forms of regimes 1–4 in all 201 saturated draws of a 270-draw audit.

Step 5: Comparative statics across the cost spectrum. In the low-cost saturation regime ($\kappa_L \leq \bar{\kappa}_L$ with $k > k_{\text{crit}}$ and $k \geq m_H^*$):

$$1 - a_H^{SB} \;=\; \frac{\frac{\kappa_L(k+m_H^*)}{2} - c_Q}{s}.$$

As κ_L increases toward $\bar{\kappa}_L$, type L's utility from completing k attributes decreases, intensifying the temptation to mimic the asking bundle. To deter mimicry, the leader must distort a_H^{SB} progressively downward (lower a_H). In the boundary relaxation regime ($\bar{\kappa}_L < \kappa_L < c_Q^{eff}/k$), the leader relieves this pressure by granting clarification to type L ($a_L = 1, m_L < k$), which raises U_L and moderates the downward distortion on type H relative to the saturation formula, dampening distortion growth without reversing its negative direction. Finally, when friction reaches or exceeds task stakes ($c_Q^{eff} \geq \Lambda$, or $\lambda_A - c_Q \geq \mu_A(\Lambda - c_Q)$), the leader prefers zero asking for all types ($\partial\Pi/\partial a = \mu_A(\Lambda - c_Q^{eff})(k - m) \leq 0$). Clarification is uniformly shut down: $a_L = a_H = 0$, and each type is left at its unassisted optimum $m_\kappa = \min\{k, \Lambda/\kappa\}$, which both parties then rank identically. When both types satisfy $\kappa \leq \Lambda/k$, they receive the identical bundle $(k, 0)$ and IC_L and IC_H hold with equality. This co-activity reflects technological clarification shutdown due to prohibitive friction, whereas genuine adverse-selection screening distortion ($a_H^{SB} < a_H^B = 1$) operates strictly in the cost-effective asking regime $c_Q < c_Q^{eff} < \Lambda$. □

B Proof of Theorem 1

Write $\bar{m}$ for $\bar{m}^*$ and $c_Q^{eff} = c_Q + \tilde{\rho} = c_Q + rs$.

Part (i). Because $U_1(\cdot, \kappa)$ peaks at c_Q/κ, Π_{std} is differentiable with $\Pi'_{\text{std}}(\bar{m}) = \int_{\kappa \geq c_Q/\bar{m}} (c_Q^{eff} - \kappa\bar{m}) f(\kappa)\, d\kappa$; the boundary of the pool contributes nothing because $m_{\bar{m}}$ is continuous in κ. At an interior maximum this derivative is zero, which is the stated formula. Since $\partial\Pi_{\text{std}}/\partial\tilde{\rho} = -\mathbb{E}[k - m_{\bar{m}}(\kappa)]$ is nondecreasing in $\bar{m}$, Π_{std} has increasing differences in $(\bar{m}, \tilde{\rho})$, and $\bar{m}^*$ is nondecreasing in $\tilde{\rho}$ by Topkis's theorem. For later use, note that substituting $c_Q^{eff} = rs + c_Q$ and integrating by parts turns the first-order condition into

$$\int_{\kappa_b}^{\bar{\kappa}} \Big[r f(t)(\Lambda - t\bar{m}) + (1 - r)F(t)\,\bar{m} \Big] dt \;=\; (1 - r)\,(\bar{\kappa}\bar{m} - c_Q), \tag{1}$$

using $\kappa_b \bar{m} = c_Q$ when $\kappa_b > \underline{\kappa}$, and $F(\underline{\kappa}) = 0$ when $\kappa_b = \underline{\kappa}$.

Part (iv). As $\tau \to \bar{\kappa}$, $T(\tau) = r f(\bar{\kappa})(\Lambda - \bar{\kappa}\bar{m})(\bar{\kappa} - \tau) + o(\bar{\kappa} - \tau)$, so $T \geq 0$ near $\bar{\kappa}$ requires $\bar{m} \leq \Lambda/\bar{\kappa}$. Let $\zeta(t)$ denote the integrand of T, so $T' = -\zeta$ and $T(\bar{\kappa}) = 0$. Subtracting $(1 - r)\bar{m}(\bar{\kappa} - \kappa_b)$ from both sides of (1) gives $T(\kappa_b) = (1 - r)(\kappa_b\bar{m} - c_Q) \geq 0$. If κ is uniform, ζ is affine in t with $\zeta(\bar{\kappa}) = r f(\bar{\kappa})(\Lambda - \bar{\kappa}\bar{m})$. If $\zeta(\bar{\kappa}) \geq 0$, then either $\zeta \geq 0$ on $[\kappa_b, \bar{\kappa}]$, so T decreases to $T(\bar{\kappa}) = 0$, or ζ changes sign once from negative to positive, so T first rises from $T(\kappa_b) \geq 0$ and then falls to 0. In both cases $T \geq 0$. Hence for uniform κ the tail condition is equivalent to $\bar{m} \leq \Lambda/\bar{\kappa}$.

Part (ii): allocation and participation. Facing the standard, a user who meets it chooses $\max\{\bar{m}, c_Q/\kappa\} \leq k$ (because $\underline{\kappa} \geq c_Q/k$), and one who does not can do no better than the unassisted payoff $U_0^*(\kappa) = U_0(m_0, \kappa)$ with $m_0 = \min\{k, \Lambda/\kappa\}$. For $\kappa < c_Q/\bar{m}$, meeting the standard gives the first best, which beats U_0^* by Proposition 3. For $\kappa \geq c_Q/\bar{m}$, we have $\kappa\bar{m} \geq c_Q$ and, by part (iv), $\bar{m} \leq \Lambda/\kappa$, so $m_0 \geq \bar{m}$ and

$$U_0(m_0, \kappa) - U_0(\bar{m}, \kappa) \;=\; \int_{\bar{m}}^{m_0} (\Lambda - \kappa t)\, dt \;\leq\; (\Lambda - \kappa\bar{m})(m_0 - \bar{m}) \;\leq\; s\,(k - \bar{m}),$$

which says $U_0^*(\kappa) \leq U_1(\bar{m}, \kappa)$. Every type therefore meets the standard. With $a = 1$ for every type, $U = U_1(m^*(\kappa), \kappa)$, and G_{m^*} is constant because $(c_Q - \kappa m^*)\, m^{*\prime} = 0$ everywhere (either $m^* = c_Q/\kappa$ or m^* is flat). So $U(\bar{\kappa}) = \min G_{m^*}$, and Lemma 5 gives $\Pi_{\text{std}}(\bar{m}) = J[m^*]$.

Part (ii): a Lagrangian bound. For any probability measure μ on $[\underline{\kappa}, \bar{\kappa}]$ with distribution function M, $\min_\kappa G_m \le \int G_m \, d\mu$, and by Fubini's theorem $\int G_m \, d\mu = \int U_1(m(\kappa), \kappa)\, \mu(d\kappa) - \int M(t)\, m(t)^2/2 \, dt$. Hence $J[m] \le \mathcal{L}(m)$, where

$$\mathcal{L}(m) \;\equiv\; \int_{\underline{\kappa}}^{\bar{\kappa}} \Big[r f(t)\, U_0(m(t), t) + (1-r)\big(F(t) - M(t)\big) \frac{m(t)^2}{2} \Big] dt \;+\; (1-r) \int U_1(m(\kappa), \kappa)\, \mu(d\kappa).$$

Let $c \equiv rs/[(1-r)c_Q]$ and $A(\tau) \equiv \int_{\kappa_b}^{\tau} [r f(t)(\Lambda - t\bar{m}) + (1-r)F(t)\bar{m}]\, dt \,/\, [(1-r)(\tau\bar{m} - c_Q)]$ for $\tau > \kappa_b$. Choose μ^* through its distribution function

$$M(t) \;=\; \begin{cases} F(t) + c\, t f(t), & \underline{\kappa} \le t < \kappa_b, \\ \max\big\{ M(\kappa_b^-),\, F(t),\, \sup_{\kappa_b < \tau \le t} A(\tau) \big\}, & \kappa_b \le t < \bar{\kappa}, \\ 1, & t = \bar{\kappa}, \end{cases}$$

with $M(\kappa_b^-) \equiv 0$ when $\kappa_b = \underline{\kappa}$. The first piece is nondecreasing because $tf(t)$ is, the second is nondecreasing by construction, and $M(\kappa_b^-) = \lim_{\tau \downarrow \kappa_b} A(\tau)$ when $\kappa_b > \underline{\kappa}$ (by l'Hôpital's rule and $\kappa_b \bar{m} = c_Q$). Subtracting from (1) shows that $A(\tau) \le 1$ if and only if $T(\tau) \ge 0$, so under the tail condition M is a distribution function. It may have atoms at $\underline{\kappa}$ and $\bar{\kappa}$.

Because $M \ge F$, the integrand of $\mathcal{L}$ is concave in $m(t)$, and so is U_1. Hence for any nonincreasing m,

$$\mathcal{L}(m) - \mathcal{L}(m^*) \;\le\; \int \big(m(t) - m^*(t)\big)\, D(dt),$$

where D is the signed measure

$$D(dt) \;\equiv\; \Big[r f(t)(\Lambda - t m^*) + (1-r)(F - M)\, m^* \Big] dt + (1-r)\,(c_Q - t\, m^*)\, \mu^*(dt).$$

Let $\Phi(t) \equiv D([\underline{\kappa}, t])$. On $[\underline{\kappa}, \kappa_b)$, $m^* = c_Q/t$, so the μ^* term vanishes and the density term equals $rfs - (1-r)\, c\, t f\, c_Q/t = 0$; hence $\Phi = 0$ there. On $[\kappa_b, \bar{\kappa}]$, $m^* = \bar{m}$, and integrating the μ^* term by parts gives

$$\Phi(\tau) \;=\; (1-r)\,(\tau\bar{m} - c_Q)\,\big[A(\tau) - M(\tau)\big] \;\le\; 0,$$

with $\Phi(\bar{\kappa}) = 0$ because $A(\bar{\kappa}) = 1$ by (1) and $M(\bar{\kappa}) = 1$. Integrating by parts once more, $\int (m - m^*)\, dD = -\int \Phi\, d(m - m^*) = -\int \Phi\, dm \le 0$: the boundary terms vanish, $\int \Phi\, dm^* = 0$ because $\Phi = 0$ wherever m^* varies, and $\Phi \le 0$ while m is nonincreasing. So $\mathcal{L}(m) \le \mathcal{L}(m^*)$ for every nonincreasing m. Finally, G_{m^*} is constant, so $\int G_{m^*}\, d\mu^* = \min G_{m^*}$ and $\mathcal{L}(m^*) = J[m^*]$. For any incentive-compatible mechanism, Lemma 5 then gives $\Pi \le J[m] \le \mathcal{L}(m) \le \mathcal{L}(m^*) = \Pi_{\text{std}}(\bar{m})$.

Part (iii). Suppose $T(\tau_0) < 0$ for some $\tau_0 \in [\kappa_b, \bar{\kappa})$. Lower effort on $[\tau_0, \bar{\kappa}]$ from $\bar{m}$ to $\bar{m} - \varepsilon$, keep m^* elsewhere, and set $U(\bar{\kappa}) = \min G_m$. Effort stays nonincreasing, so the mechanism is incentive compatible, and for small ε it satisfies $0 \le x \le s(k - m)$. The first term of J changes by $-\varepsilon \int_{\tau_0}^{\bar{\kappa}} [r f(\Lambda - t\bar{m}) + (1-r)F\bar{m}]\, dt + O(\varepsilon^2)$. For $\kappa < \tau_0$, $G_m(\kappa)$ rises by $\varepsilon\bar{m}(\bar{\kappa} - \tau_0) + O(\varepsilon^2)$; for $\kappa \ge \tau_0$ it rises by $\varepsilon(\bar{\kappa}\bar{m} - c_Q) + O(\varepsilon^2)$, which is at least as much because $\tau_0 \bar{m} \ge c_Q$. So $\min G_m$ rises by $\varepsilon\bar{m}(\bar{\kappa} - \tau_0) + O(\varepsilon^2)$, and J changes by $-\varepsilon\, T(\tau_0) + O(\varepsilon^2) > 0$. By Lemma 5 this is the payoff of the perturbed mechanism, which therefore beats the standard. The tersest users in it receive $a < 1$. □

C Unclipped versus Clipped Pooling Payoffs

The qualitative finding of Corollary 4, that unbiased pooling is a corner solution rather than an interior compromise, holds for every type distribution, with or without clipping (by the convexity argument given there). The unclipped corner difference formula $\Pi(1)-\Pi(0) = (\Lambda - c_Q)[k - k^*]$ relies on unconstrained best responses. Table 1 documents this unconstrained boundary across varying task sizes k and baseline guessing accuracies g, holding $L = 10, c_Q = 2$, and $\kappa \sim \text{Uniform}[0.2, 0.6]$ (where $\mathbb{E}[1/\kappa] \approx 2.747$).

g	$\Lambda = L(1-g)$	Threshold k^*	Task Size k	Unclipped $\Delta\Pi$	Clipped $\Delta\Pi$
0.30	7.0	12.36	5	−36.80	+0.15
0.30	7.0	12.36	10	−11.80	+5.49
0.30	7.0	12.36	15	+13.20	+20.10
0.50	5.0	9.61	5	−13.84	+0.15
0.50	5.0	9.61	9	−1.84	+3.69
0.50	5.0	9.61	10 (Benchmark)	+1.16	+5.42
0.50	5.0	9.61	12	+7.16	+9.62
0.50	5.0	9.61	15	+16.16	+17.12
0.70	3.0	6.87	5	−1.87	+0.15
0.70	3.0	6.87	8	+1.13	+1.71
0.70	3.0	6.87	10	+3.13	+3.32

Table 1: Sensitivity of the unbiased pooling corner choice across task complexity k and guessing capability g, for $L = 10, c_Q = 2, \kappa \sim \text{Uniform}[0.2, 0.6]$, where $\Delta\Pi \equiv \Pi(1) - \Pi(0)$. Columns compare the unconstrained closed form $(\Lambda - c_Q)[k - k^*]$ against the physically feasible clipped payoff difference computed by numerical integration. In the unclipped algebra, guessing appears to win whenever $k < k^*$ (e.g., −36.80 at $g = 0.30, k = 5$), but when followers are constrained to feasible effort $m \leq k$, asking strictly dominates across all rows (+0.15 at $g = 0.30, k = 5$), confirming Proposition 3. While binary symmetric attributes naturally correspond to baseline guessing accuracy $g = 0.50$, configurations with $g = 0.30$ and $g = 0.70$ represent tasks with non-binary attribute domains (such as multi-choice syntactical options where random guessing yields $1/M < 0.5$, or informed priors where guessing yields > 0.5).

As shown in Table 1, once followers are restricted to feasible specification effort $m \in [0, k]$, asking ($a^{SE} = 1$) weakly dominates guessing across all 11 configurations under linear risk. The apparent dominance of guessing in the unclipped column stems entirely from low-cost types choosing unfeasible specification levels $m = \Lambda/\kappa > k$. At $k = 5$, for all $\kappa \in [0.2, 0.6]$, the unassisted specification choice under guessing is $\min\{5, \Lambda/\kappa\} = 5 = k$ across all $g \in \{0.30, 0.50, 0.70\}$, because $\Lambda \geq 3.0$ and $\kappa \leq 0.6$. When followers fully saturate specification effort under silence, the task risk parameter Λ drops out completely from the comparison, and the net gain from asking $\Delta\Pi = \mathbb{E}[U(m^*(1); 1) - U(k; 0)] = +0.15$ depends strictly on $c_Q = 2$ and κ. Under the exact conjunctive payoff $Vq(a, g)^{k-m}$, guessing can beat asking for a given type (Remark 5), but in our scans this happens only for terse types on long tasks (Section 12).

D Replication

Every numerical statement in the paper is produced by the companion code. The script `run_all_math_checks.py` runs all checks and writes tables, figures, and reports to `outputs/`; the

main scripts, in `underspec_sim/verifications/`, are as follows.

- Propositions 3–4 and Corollaries 2–4: `verify_prop3.py`, `verify_prop4.py`, `verify_cor2.py`, `verify_cor3.py`, `verify_mu_lambda_corner.py`; Table 1: `generate_table1.py`.
- Proposition 5 and the trap's incidence: `verify_prop5_guessing_trap.py`, `verify_guessing_trap_incidence.py`.
- Propositions 6–7 and Lemma 3: `verify_prop5.py`, `verify_prop6.py`, and the exact grid audit `verify_prop7_regimes.py`; Lemma 4: `verify_continuous_implementation.py`.
- Corollaries 5–6, Remark 12, and Figures 4–5: `compare_regimes_runner.py`, `verify_welfare_ordering.py`, `sweep_distortion.py`; Proposition 8: `verify_heterogeneous_stakes.py`.
- Section 8: `verify_continuum_standard.py`, which solves the optimal mechanism for up to 80 types and compares it with Theorem 1.
- Section 12: `verify_exact_conjunctive.py`, `verify_exact_firstbest_gap.py`, `verify_exact_bias_sweep.py`, `verify_assumption1.py`, `verify_monotonicity_survival.py`.

The simulated-user harness for the experimental protocol of Section 11 is in `underspec_sim/experiments_llm`.